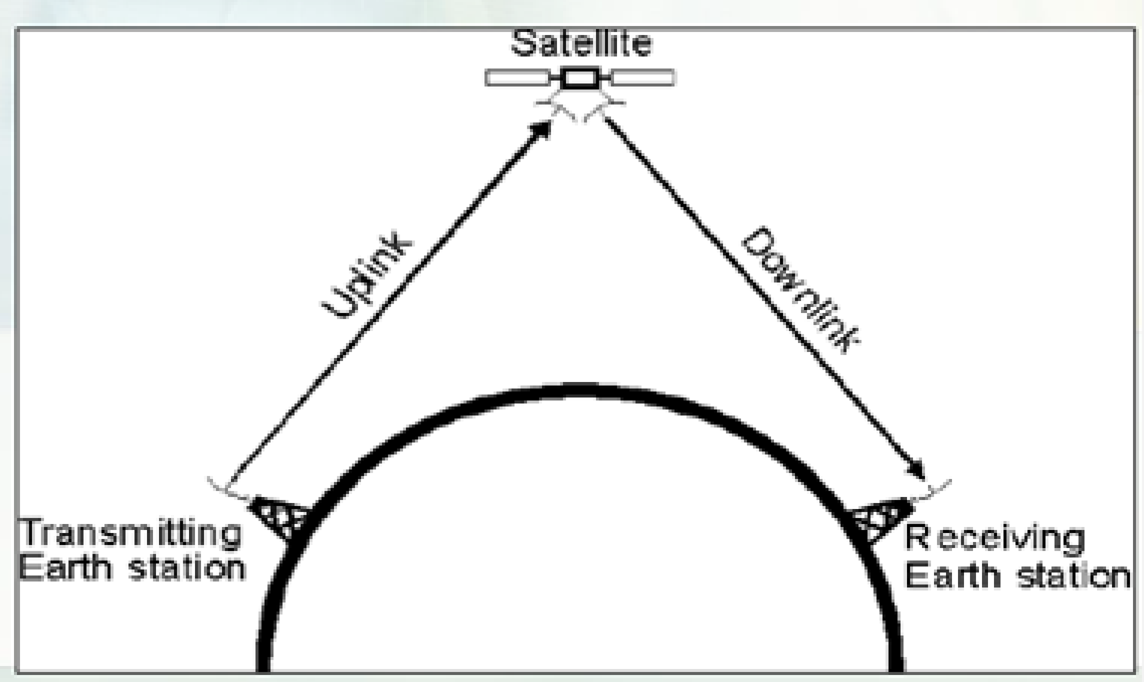
KNUST **RANKS NO.1 GLOBALLY** FOR THE PROVISION OF QUALITY EDUCATION **(SDG 4)** By: THE WORLD UNIVERSITY RANKINGS

TE 456: Satellite Communication and Navigation System

Elements of SatCom and 5G Systems

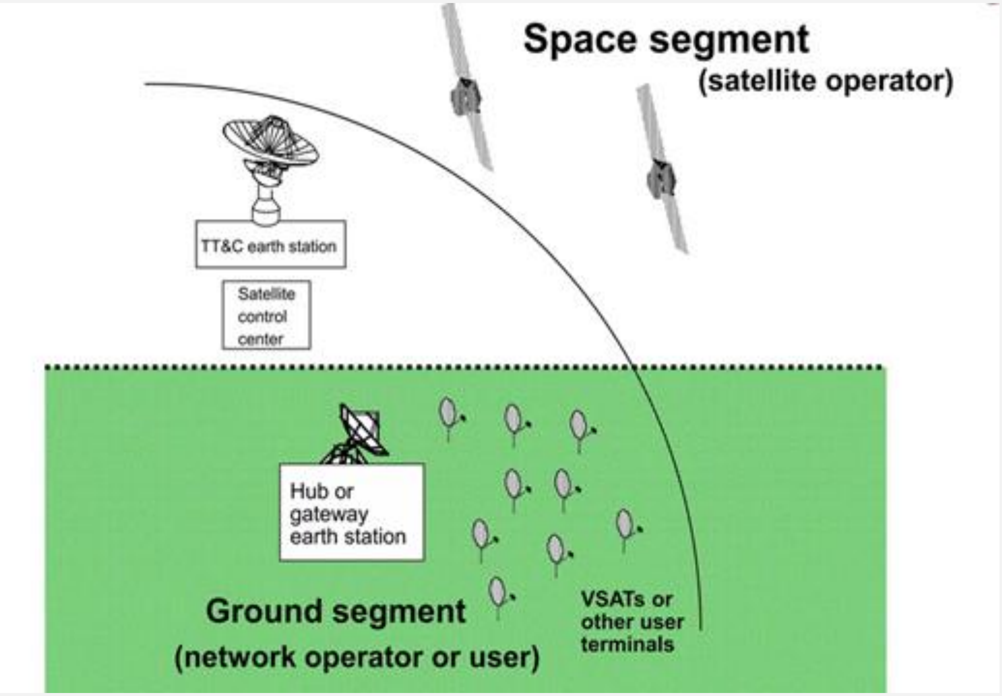
Communication Satellite

- A communication satellite acts as a repeater



Satellite Subsystems

- A satellite communication system consists of
  1. Earth segment
  2. Space segment
     - both are divided into distinct subsystems



- Space Segment Subsystems
  a. Communications payload
  b. Space platform (bus)
- Earth Segment Subsystems
  a. Ground station/GW
  b. User terminals

Communication Subsystem

- A communication satellite
  - exists to provide a platform in orbit for the relaying of data/signal
    - All other subsystems on the satellite exist solely to support the communications system
- A repeater is simply
  - a receiver linked to a transmitter, always using different radio frequencies,
    - that can receive a signal from one *earth station*, amplify it, and retransmit it to another earth station

Space Segment Subsystems

- The communications subsystem is
  - responsible for **ensuring telecommunication between the satellite and another system**, which may be either another satellite or a ground station
- Through this subsystem
  - The following data must pass from the spacecraft to operators and users
    a. Payload mission data
    b. spacecraft housekeeping data

Communications Payload

- The payload
  - consists of two distinct parts with well-defined interfaces

    i. repeaters/transponders
      - amplify and frequency-translate the received uplink signals (e.g., from an Earth station) so they can be re-transmitted on a downlink frequency without interference

    ii. Antennas
      - Receive incoming signals from Earth and transmit the processed signals back down, directing beams to specific coverage areas (footprints)

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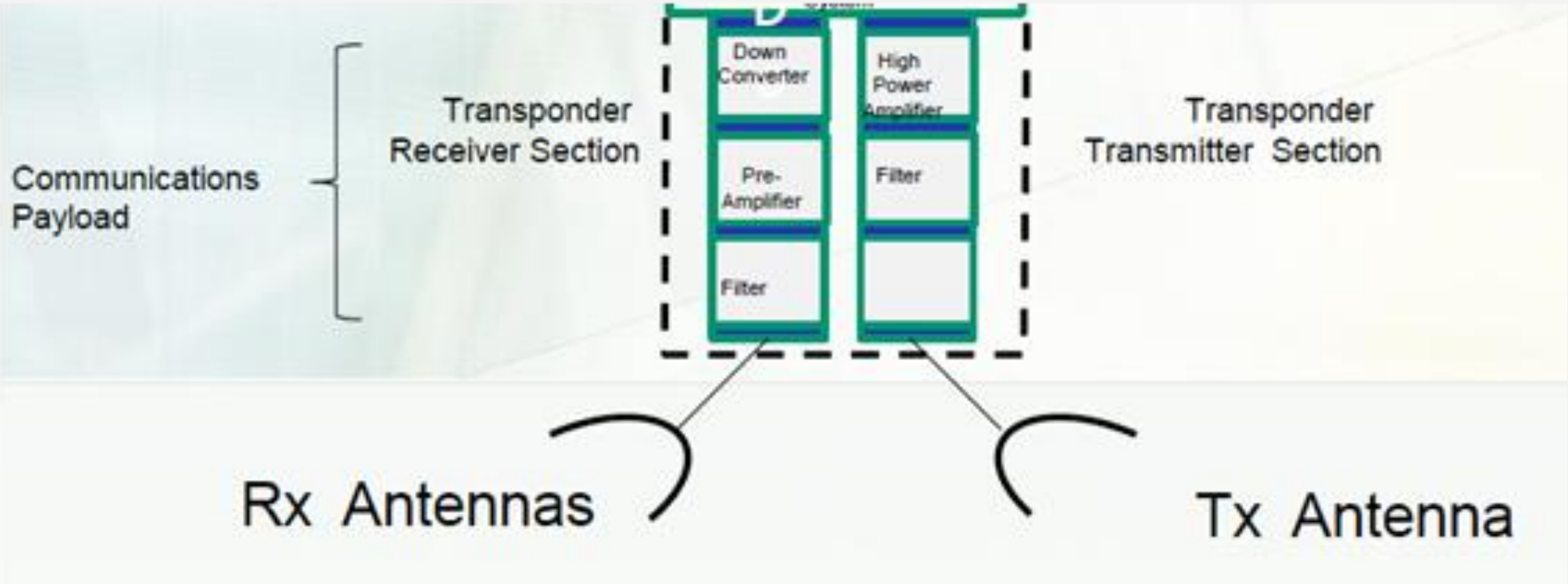
Communications Payload

- Repeater and Antennas



Communications Payload

- The payload
  - is the heart of the satellite, and it is responsible for performing some key functions
- Key payload functions
  1. To capture the carriers transmitted in a given frequency band by the earth stations of the network
  2. To capture as little interference as possible
  3. To amplify the received carriers while limiting noise and distortion as much as possible

Communications Payload

- Key payload functions

4. To change the frequency of the carriers received on the uplinks to that on the downlinks
    - for example, from 14 to 11GHz for Ku band and from 30 to 20GHz for Ka band
5. To provide the power required in a given frequency band at the interface with the transmitting antenna
6. To radiate the carriers in a given frequency band
7. To route the carriers from any given uplink beam to any downlink beam
8. To demodulate and remodulate carriers

Communications Payload

- Repeater/Transponder
  - It is the electronic equipment that performs a range of functions on the signals (known as *carriers*) from the receiving antenna before delivering them to the transmitting antenna
- The repeater
  - usually encompasses several channels (also called *transponders*) that are individually dedicated to sub-bands within the overall payload frequency band
    - For example, a payload bandwidth of 500MHz is divided up into channels, often 36 MHz per transponder, thus allowing about 12 transponders

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Communications Payload

- Repeater/Transponder
  - It is the electronic equipment that
    - provides the connecting link between the satellite’s
      - receive and transmit antennas
    - Performs a range of functions on the signals (known as *carriers*) from the receiving antenna before delivering them to the transmitting antenna

Communications Payload

- Repeater/Transponders
  - The architecture differs for single-beam transparent repeater, regenerative repeater, and multibeam payload

Communications Payload

- Repeater Architectures
  - Satellite repeater architectures define
    - how orbital payloads process and relay signals.
- The two primary architectures are
  - **Transparent (Bent-Pipe)**
    - which simply amplifies and frequency-shifts an uplinked signal, and
  - **Regenerative (On-Board Processing)**,
    - which demodulates, processes, and re-modulates the signal in space
- Emerging Architectures
  - Software defined

Communications Payload

- **Transparent (bent-pipe) Repeater**
  - Acts as a simple, high-altitude relay station
    - It receives signals from the ground, amplifies them, translates them to a new frequency, and retransmits them
  - consists of the following
    a. Low noise amplifier (LNA)
    b. A mixer associated with a local oscillator for frequency conversion
    c. A high power amplifier (HPA), which amplifies the signal after conversion

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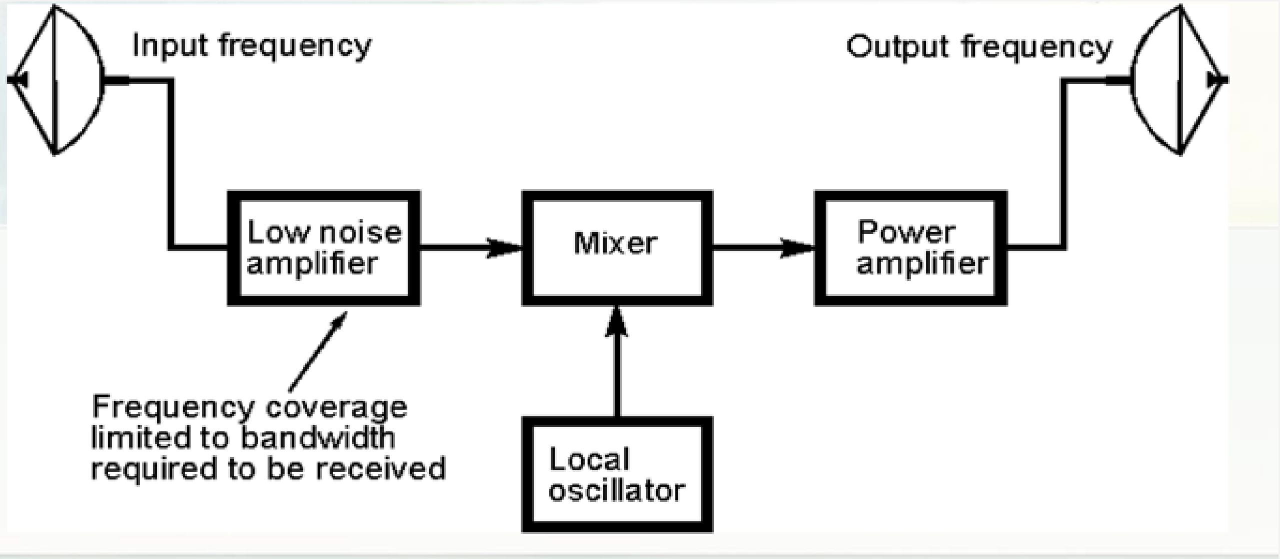
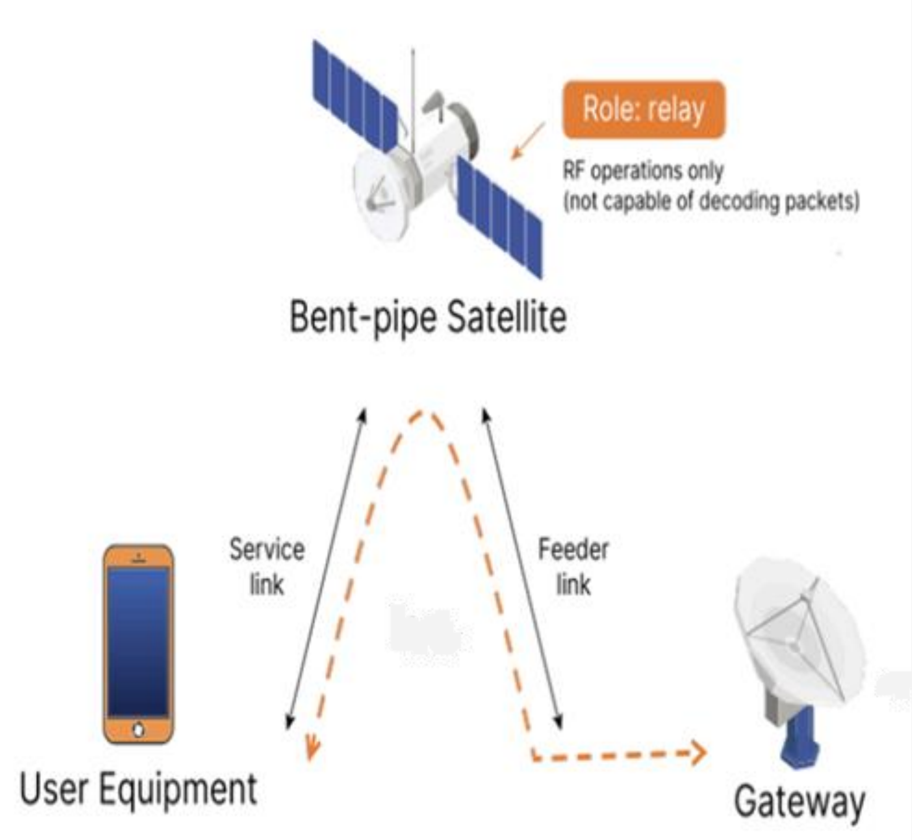
Communications Payload

- **Transparent (bent-pipe) Repeater**





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Communications Payload

- **Bent pipe**
  - It captures signals from Earth and redirects them back without demodulation
- **Signal Processing**
  - Before retransmission
    - The received uplink signal undergoes frequency conversion to the downlink frequency, after which it is amplified and then filtered

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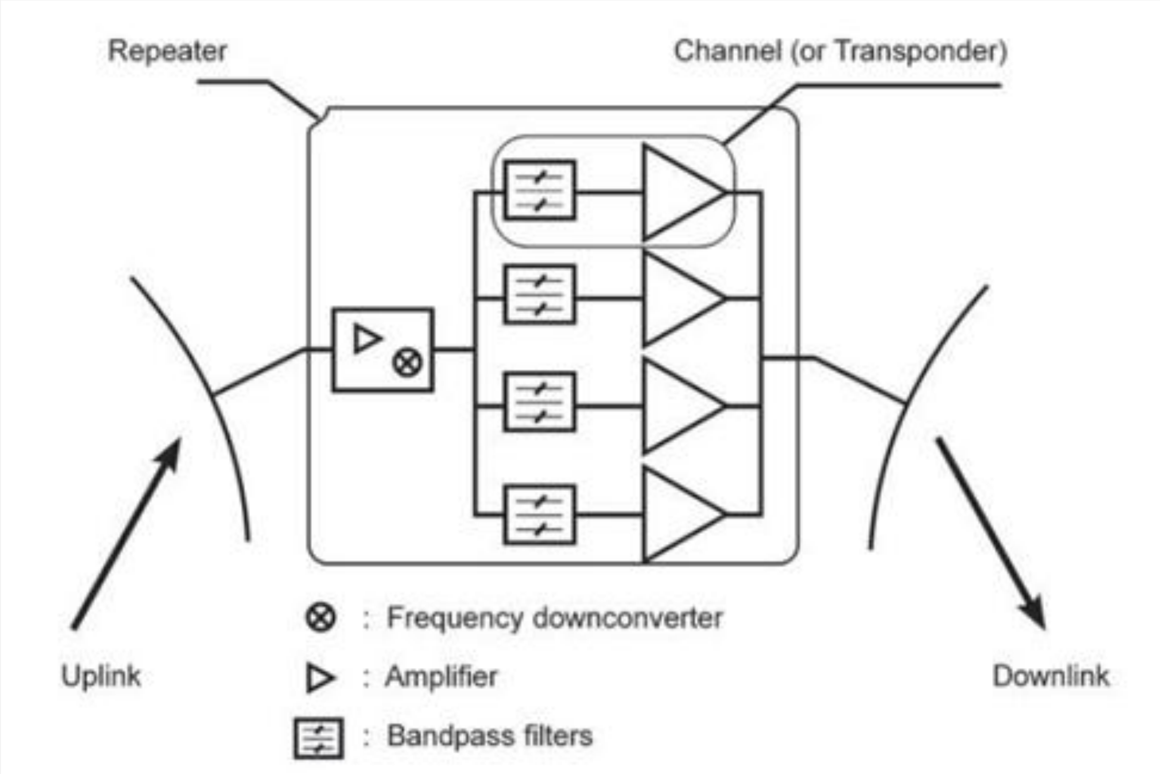
Communications Payload

- **Transparent (bent-pipe) Repeater**

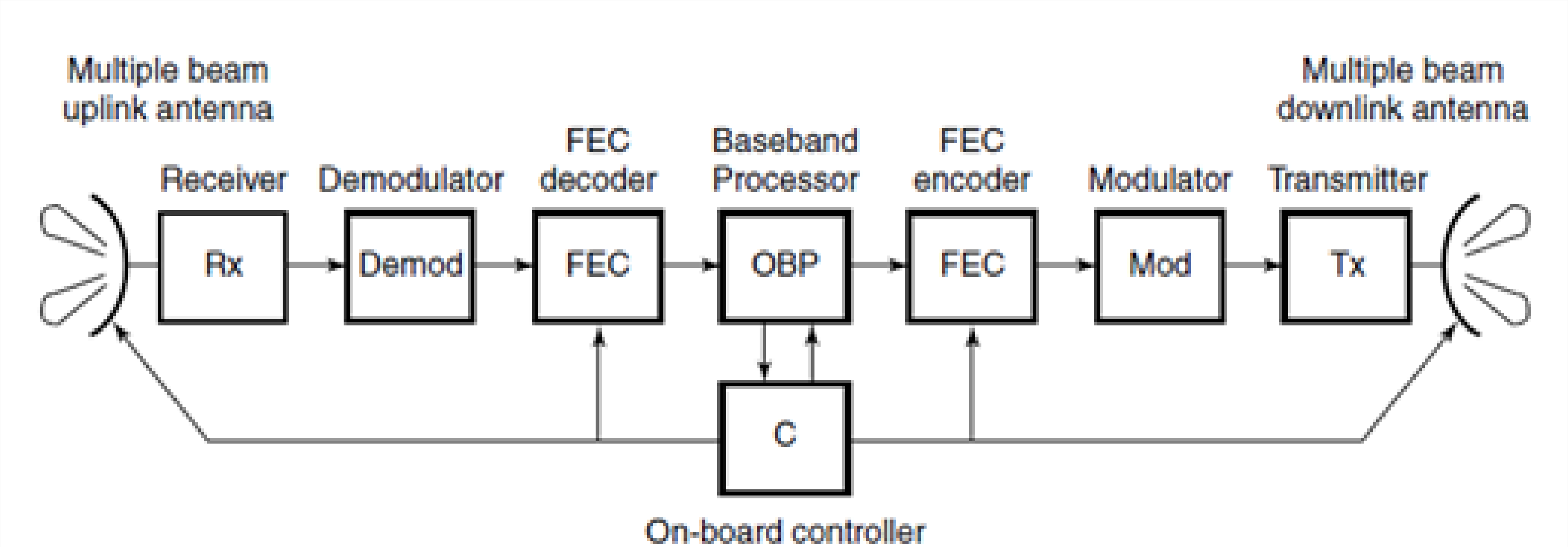


Communications Payload

- **Regenerative repeater** (processing transponder)
    - is an on-board satellite system that fully demodulates
        - an incoming uplink signal into baseband data, processes and routes it, and then
            - remodulates and amplifies it for the downlink
                - Unlike traditional "bent-pipe" (analog) repeaters that simply amplify and forward everything—including noise—regenerative repeaters "clean" the signal by decoding, correcting errors, and retransmitting a flawless copy

Communications Payload

- Regenerative (OBP)



Communications Payload

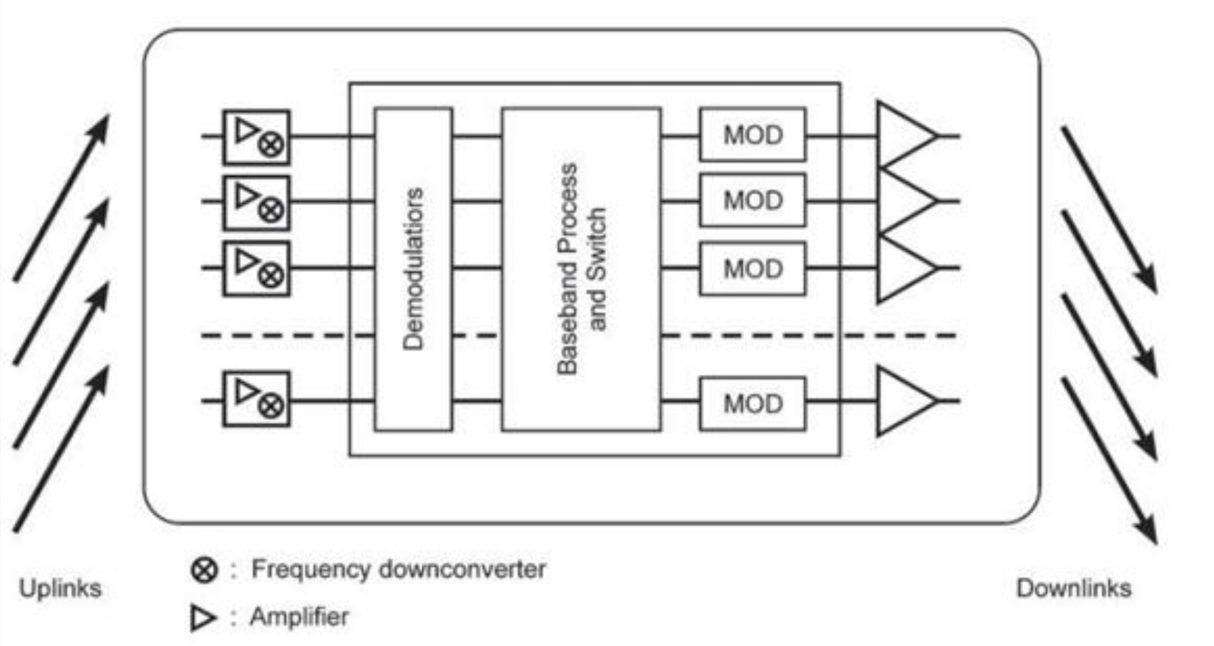
- The regenerative process broadly follows these steps
  - **Demodulation:**
    - The received RF (Radio Frequency) signal is amplified, down-converted, and demodulated to recover the original digital bitstream (baseband signal)
  - **Decoding and Reshaping**
    - The signal is equalized to remove distortion, and error-correction decoding (like FEC) is applied to fix any bit errors caused during the uplink
  - **Routing/Switching:**
    - The clean data is directed to the appropriate beam or Inter-Satellite Link (ISL)
  - **Remodulation:**
    - The clean, processed data is then modulated onto a new carrier frequency and power-amplified for the downlink

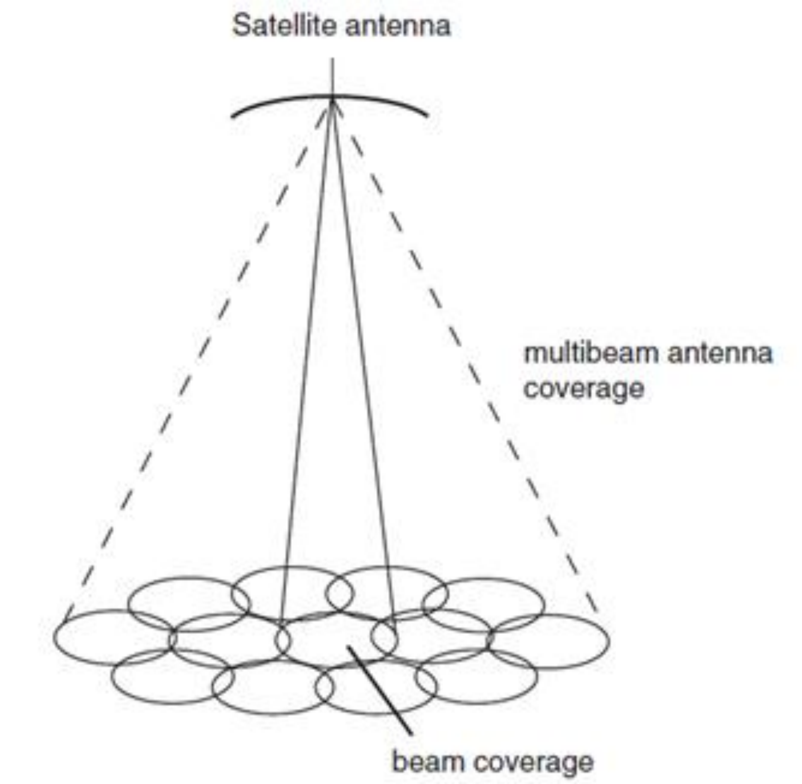
Communications Payload

- **Regenerative Repeater**
  - Like transparent repeaters, it includes uplink/downlink antennas, low-noise amplifiers (LNAs), frequency converters, and high-power amplifiers (PAs)
  - **Demodulation & Re-modulation:**
    - Unlike transparent repeaters, a regenerative repeater
      - demodulates the received uplink RF signal to recover the baseband signal.
        - Then, it re-modulates the baseband signal to produce the downlink RF signal
  - **Onboard Processing Advantages:**
    - This allows for onboard processing based on different applications and switching in the baseband

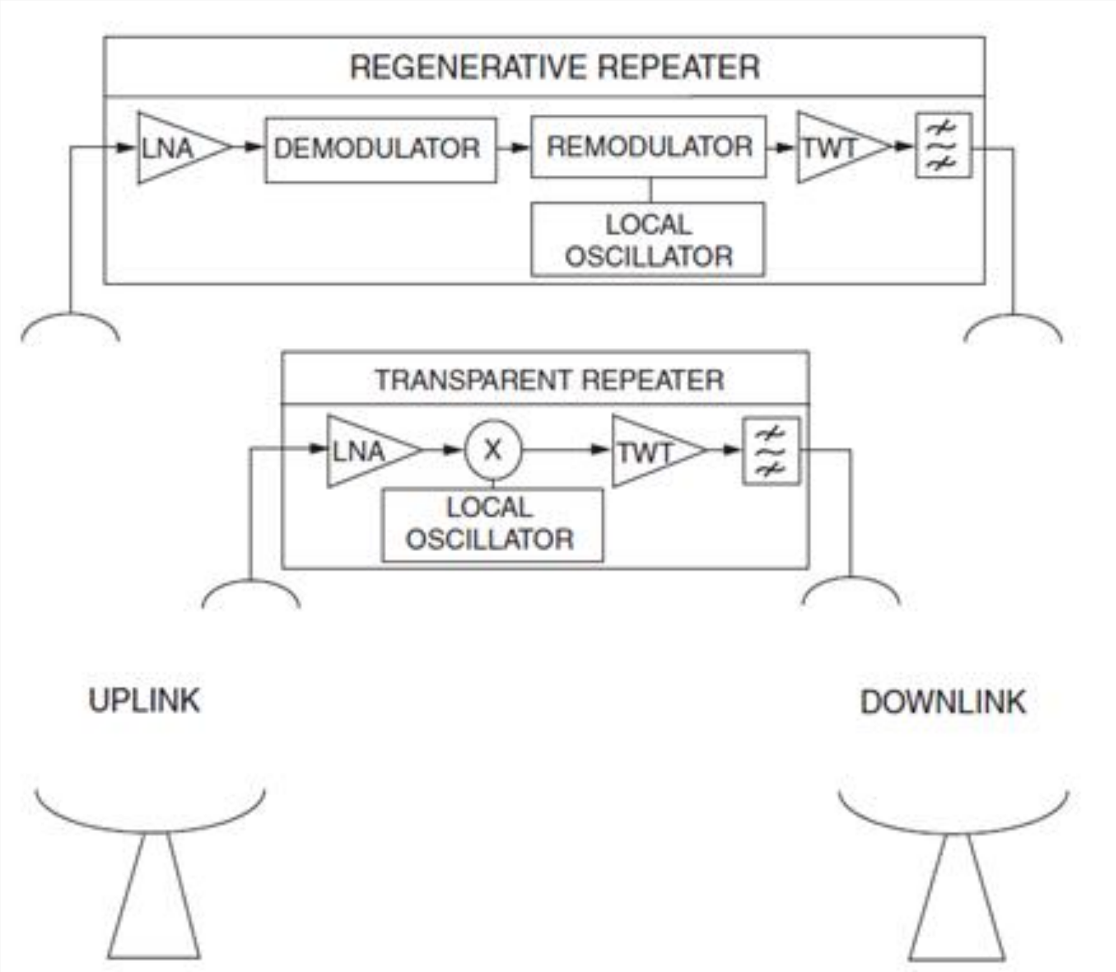
Communications Payload

- Regenerative Repeater





Communications Payload



Space Platform (Bus)

- These subsystems support the communication payload
  - and ensure the satellite remains operational in the harsh space environment
    a. Electrical Power Subsystem (EPS):
    b. Attitude and Orbit Control System (AOCS):
    c. Propulsion Subsystem:
    d. Thermal Control Subsystem
    e. Telemetry, Tracking, and Command (TTC&M):
      - A communication subsystem, which provides vital communication between the satellite and the ground control station.
        - Thus, monitors the health of all onboard subsystems, tracks the satellite's exact position, and executes commands sent from ground control

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The Earth/Ground Segment

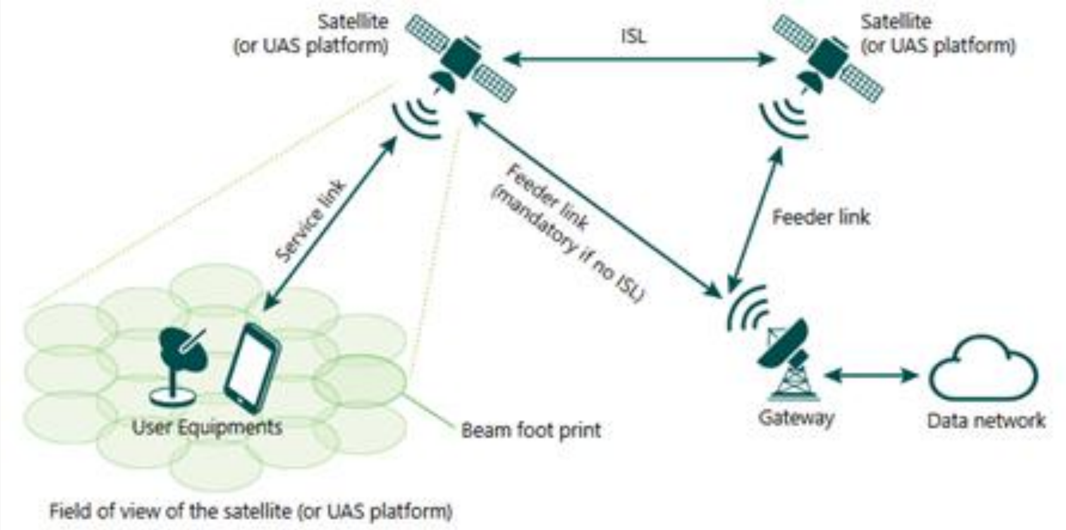
- This system includes all
  - the terrestrial stations used to control the satellite and communicate through it
    - **Ground Stations:**
      - Transmit and receive signals, monitor satellite telemetry, and send uplink commands to adjust the satellite's operations.
    - **User Terminals:**
      - The end-user equipment (e.g., satellite dishes, mobile satellite phones) used to access the communication services relayed by the satellite

Satellite Links

- A link between transmitting equipment and receiving equipment
  - consists of a radio or optical modulated carrier
- The three main links are
  1. **Service Link**
     - between the user device and the satellite
  2. **Feeder Link**
     - between the satellite and an Earth ground station/gateway
  3. **Intersatellite Link**
     - a direct communications connection between two artificial satellites in orbit

Satellite Links

- Uplinks and downlinks consist of
  - radio frequency modulated carriers
- ISLs can be either
  - radio frequency or optical
- Some large-capacity data-relay satellites
  - also use optical links with their ground stations



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Satellite Link Design

- The performance of the
  1. transmitting equipment
     - is measured by its effective isotropic radiated power (EIRP)
       - which is the power fed to the antenna multiplied by the gain of the antenna in the considered direction
  2. receiving equipment
     - is measured by G/T, the ratio of the antenna receive gain, G, in the considered direction and the system noise temperature, T
       - G/T is called the receiver's *figure of merit*
  3. link
     - can be measured by the ratio of the received carrier power, C, to the noise power spectral density, N_0
       - denoted as the C/N_0

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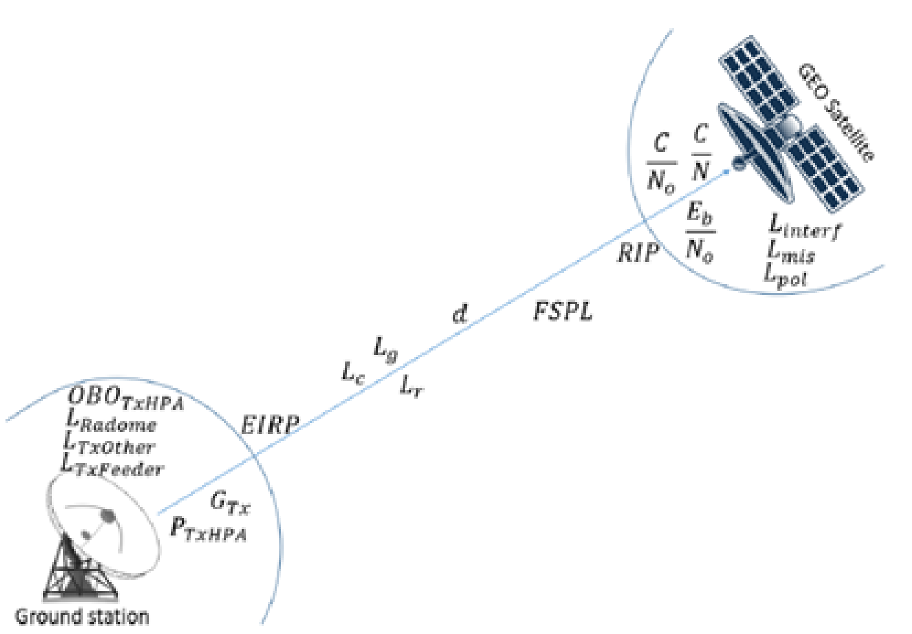
Satellite Link Budget

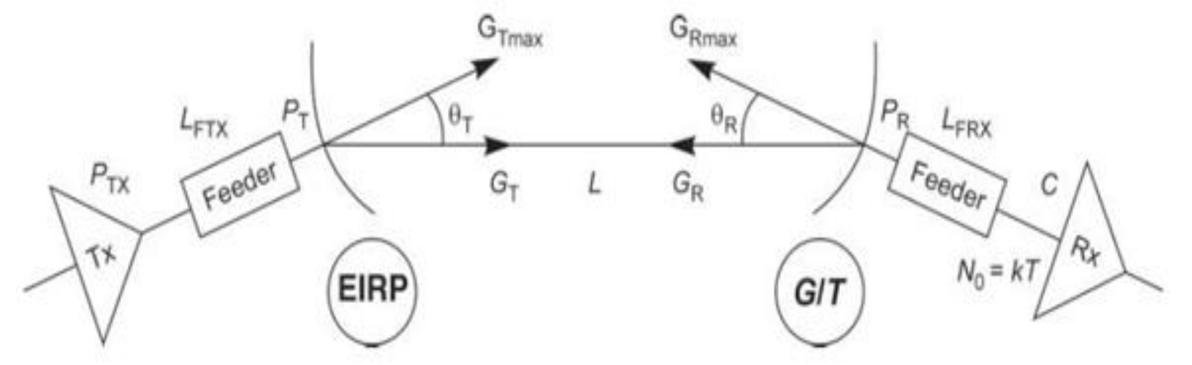
- A **satellite link budget**
  - is a structured way to predict whether a radio link will work
    - Thus,
      - how strong the signal will be when it arrives
      - how much noise it will compete with
      - how much **margin** you have in real conditions
    - It's called a "budget" because you're accounting for every gain and loss from transmitter to receiver
  - answers a basic question:
    - When the signal gets there, will it be good enough?

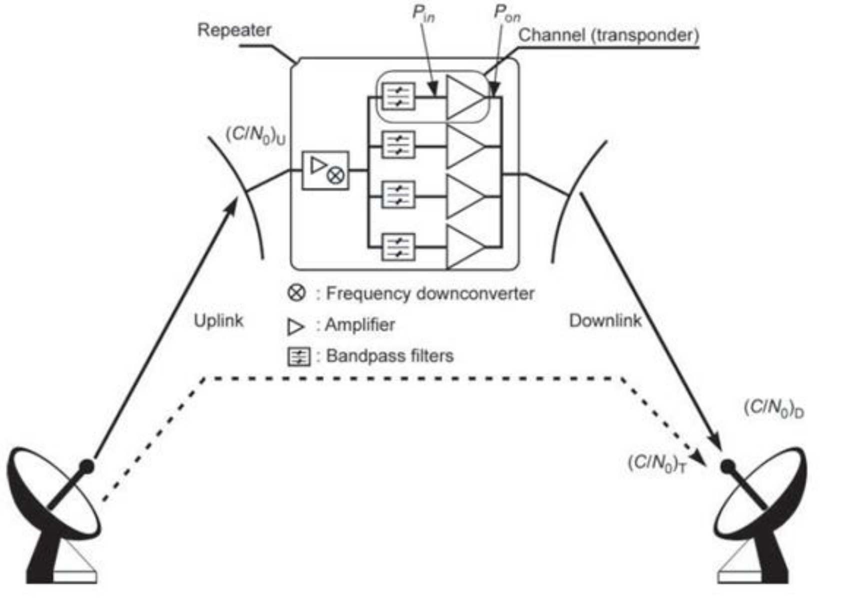
Satellite Link Budget

- The parameters
  - involved in a link budget analysis can be divided into three categories:
    1. Emission parameters
       - EIRP is the fundamental emission parameter
    2. Link parameters
       - given by the link distance and the totality of the losses that may occur related to the propagation medium
         - Free space path loss
         - rain, gas, fog/cloud attenuation
    3. Receiving parameters
       - Polarization loss
       - Antenna mis-pointing
       - System noise

Satellite Link Budget







Satellite Link Budget

- Relevant steps
  1. Calculate EIRP
  2. Add Free-Space Path Loss
  3. Add Atmospheric and Pointing Losses
  4. Add Receiver Gain and G/T
  5. Calculate Carrier-to-Noise (C/N)
  6. Convert to Eb/N_0 or C/N_0
  7. Add Fade Margin and Availability

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Satellite Link Budget

- EIRP

$$EIRP\ (\text{dBm}) = P_t\ (\text{dBm}) + G_t\ (\text{dBi}) - L_f - L_p\ (\text{dB})$$

- The receiver input power is

$$P_R = EIRP + G_R - L_{FS} - L_{ta} - L_{ra} - A_{AG} - A_{rain} - L_{POL} - L_{point}$$

- Define carrier power as the receiver input power

$$C = P_R$$

- Define noise spectral density as
  - the product of the Boltzmann constant k and system noise temperature T

$$N_0 = k \times T$$

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Satellite Link Budget

- Considering the characteristics of the receiver, such as
  - system temperature, modulation, bit rate, and interference
- P_R can be used to calculate the
  a. carrier-to-noise ratio (C/N)
  b. carrier-to-noise plus interference ratio (C/(N + I))
  c. carrier-to-noise density ratio (C/N_0)
  d. and bit energy-to-noise density ratio (Eb/N_0)

$$N = k_B T B_w$$

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Satellite Link Budget

- Define noise spectral density as
  - the product of the Boltzmann constant *k* and system noise temperature *T*

    $N_0 = k \times T$

- Calculate C/N_o

$$\frac{C}{N_0} = \frac{P_R}{kT}$$

- Received $E_b/N_o = C/N_o - 10\log_{10}(\text{user bit rate}) - 60$

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EQUATION

$$P_R = EIRP + G_R - L_{FS} - L_{ta} - L_{ra} - A_{AG} - A_{rain} - L_{POL} - L_{point}$$

where:

- $EIRP$ = Effective Isotropic Radiated Power
- G_R = Receiving antenna gain
- L_{FS} = Free space loss (Path loss)
- L_{ta} = Losses associated with transmitting antenna
- L_{ra} = Losses associated with receiving antenna
- A_{AG} = Attenuation by the atmosphere and ionosphere
- A_{rain} = Attenuation due to precipitations and clouds
- L_{POL} = Losses caused by polarization mismatch between the transmitting and the receiving antenna
- L_{point} = Losses caused by antenna depointing

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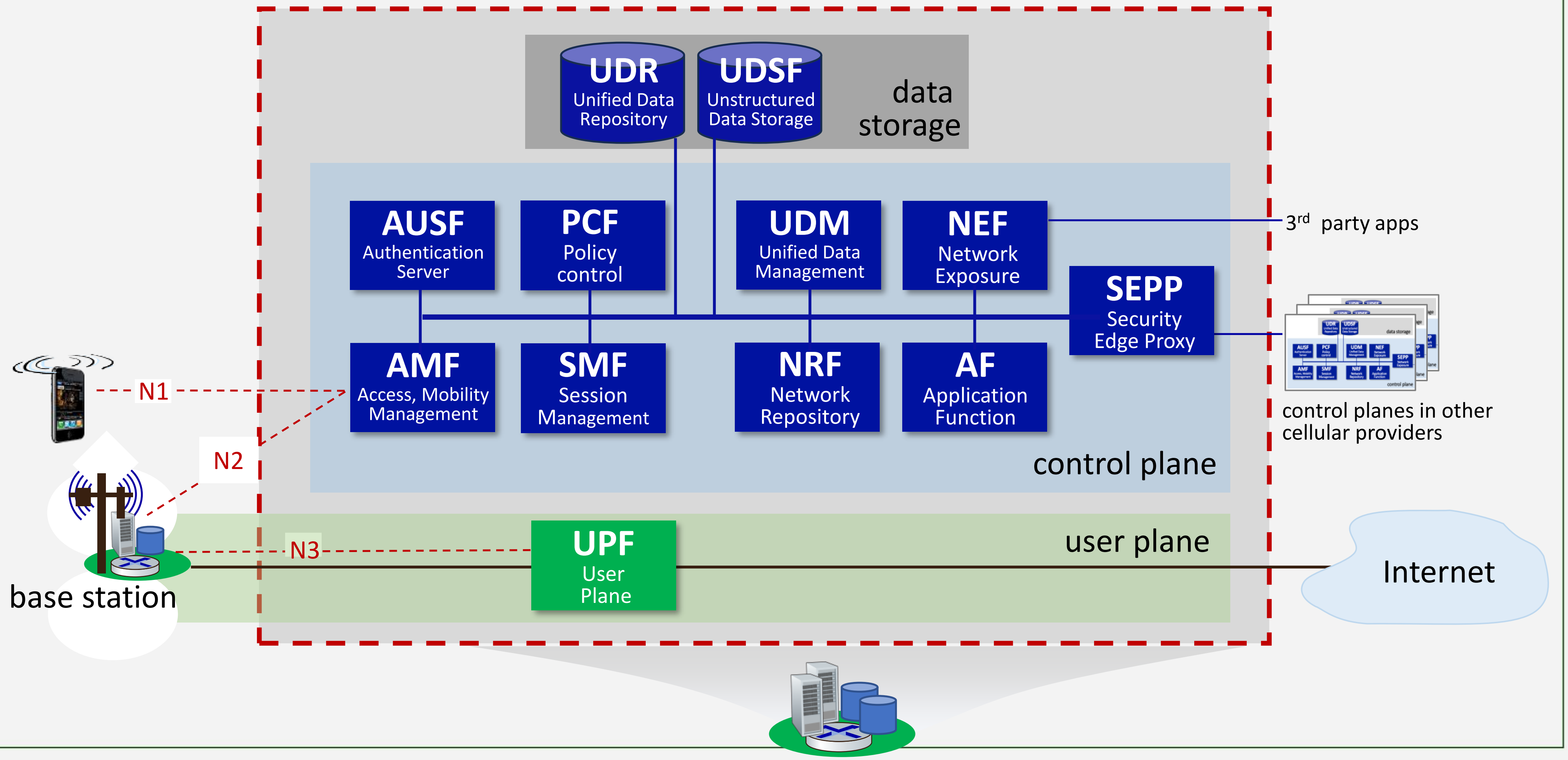
5G System

- The 5GS includes
  a. the UE
  b. the 5G radio access network RAN)
  c. the 5G Core Network (5GC)
- 5G defines
  - a service-based architecture (SBA), where entities called network functions (NFs) provide services through service-based interfaces
- Before diving into
  - The 5G RAN itself, let's situate the RAN in the context of the larger 5G cellular network

SBA



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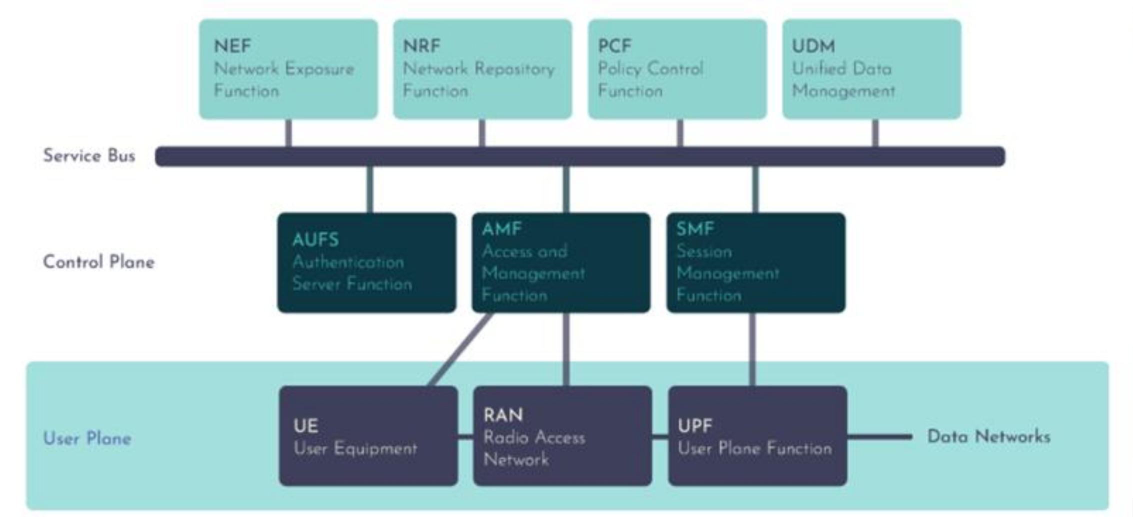
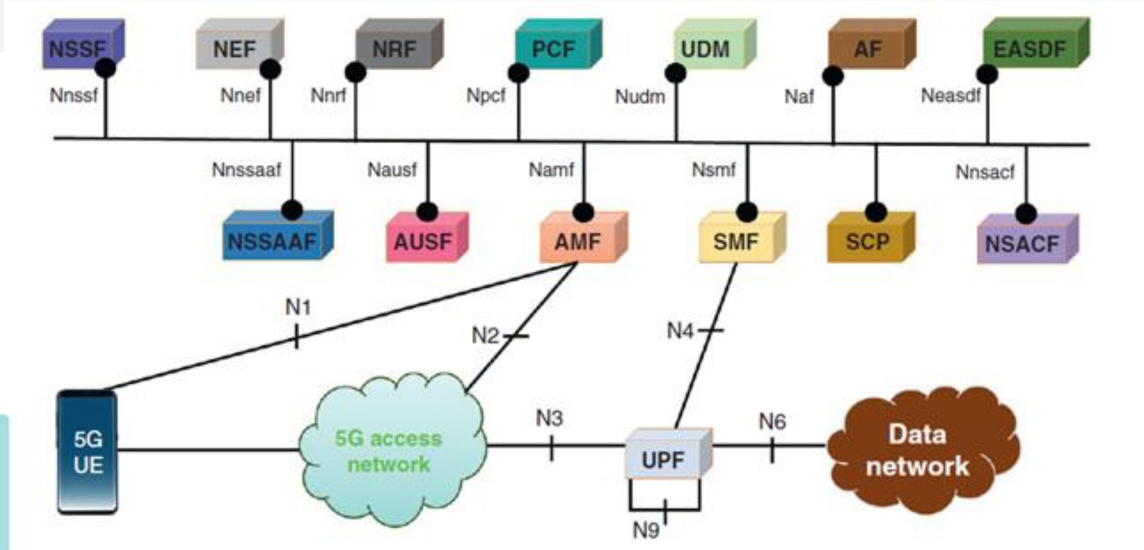
5G System Architecture

- Service-based architecture





AF: Application Function	AMF: Access & Mobility Management Function	AUSF: Authentication Server Function
CHF: Charging Function	NEF: Network Exposure Function	NRF: Network Repository Function
PCF: Policy Control Function	RAN: Radio Access Network	SMF: Session Management Function
UDM: Unified Data Management	UE: User Equipment	UPF: User Plane Function

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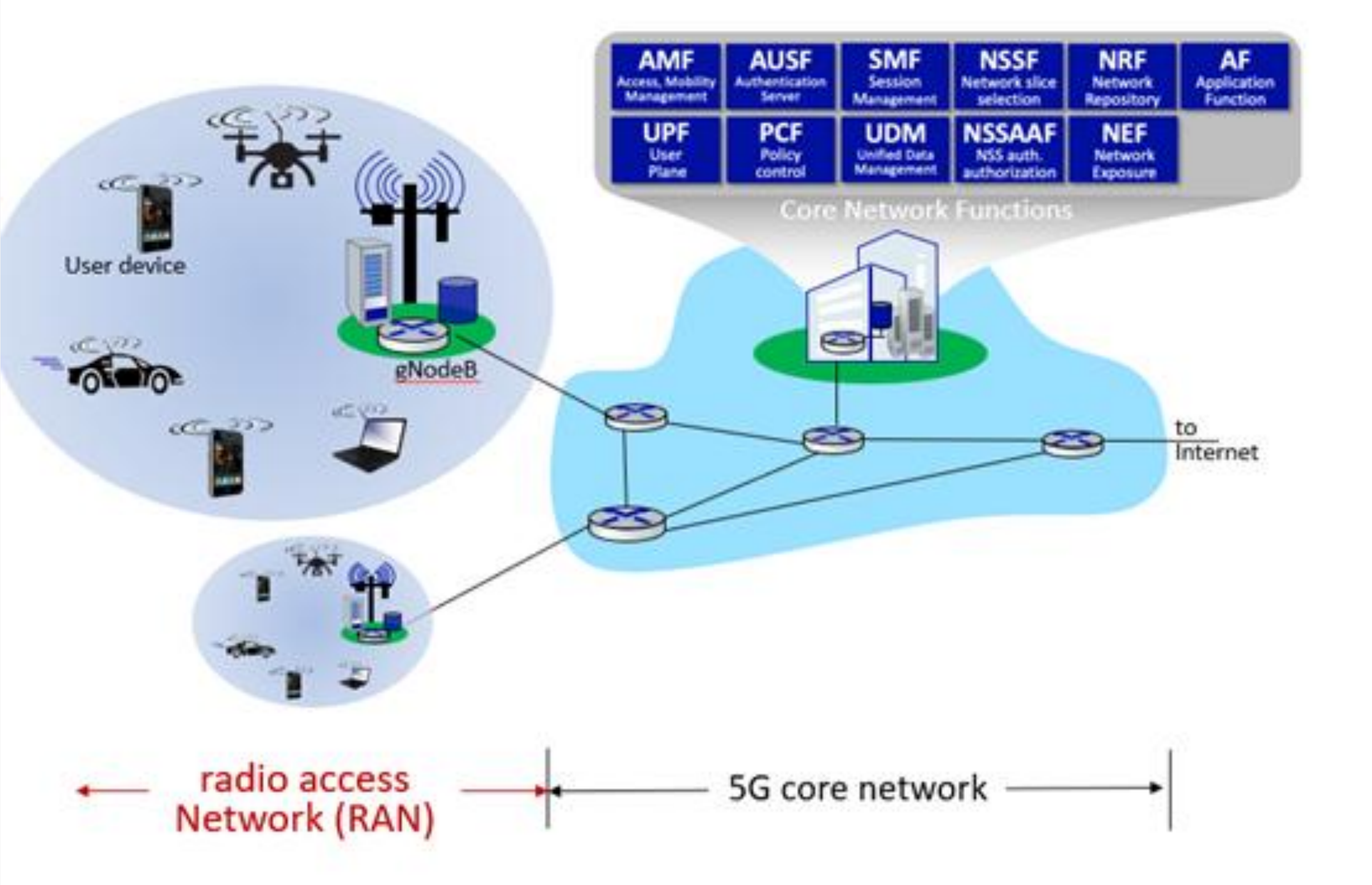
5G Architecture

- Elements of 5G



5G Radio Access Network (RAN)

- A **5G Radio Access Network (RAN)**
  - is the critical infrastructure that connects
    - smartphones and devices to the mobile core network
      - It utilizes **5G New Radio (NR)** air interfaces to convert digital data into radio frequencies, enabling ultra-high-speed, low-latency, and high-capacity wireless communication
  - It processes data in a step-by-step pipeline
    - Thus, it takes user data packets and turns them into radio waves for transmission
  - is responsible for transmitting and receiving wireless signals
    - managing radio resources, and facilitating data transfer between user devices and the core network

5G Radio Access Network (RAN)

- A **5G New Radio**
  - We need to have a good understanding of how information is communicated over NR
    i. Frequency bands
    ii. Sharing the wireless channel
    iii. PHY Frame structure
    iv. Physical channels and signals
    v. Radio Protocol Stack
    vi. 5G initial access procedure

5G NR

- Frequency bands
  - Frequency Range 1 (FR1) [410MHz – 7.125 GHz]
  - Frequency Range 2 (FR2) [24 – 52GHz]

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5G spectrum: three spectrum bands

High band frequencies: 25–66 GHz range (aka mmwave)

- 26 GHz, 40 GHz, 50 GHz, 66 GHz bands popular
- short distances (< mile), high speeds (<3 Gbps)
- line of sight transmission: poor penetration of trees, buildings, ...

5G

Mid-band frequencies: 1 – 7.125 GHz ranges

- balance distances (~5 miles ?) and transmission rates (100–900 Mbps)
- 1.8, 3.3 GHz to 3.8 GHz, 6 GHz bands popular

3.4 - 6 GHz: 5G 4G

1-2.6 GHz: 5G 4G 3G 2G

Low band frequencies: (< 1 GHz range)

- covers longer distances (10's of miles), but at lower speeds (50–250 Mbps)

5G 4G 3G 2G

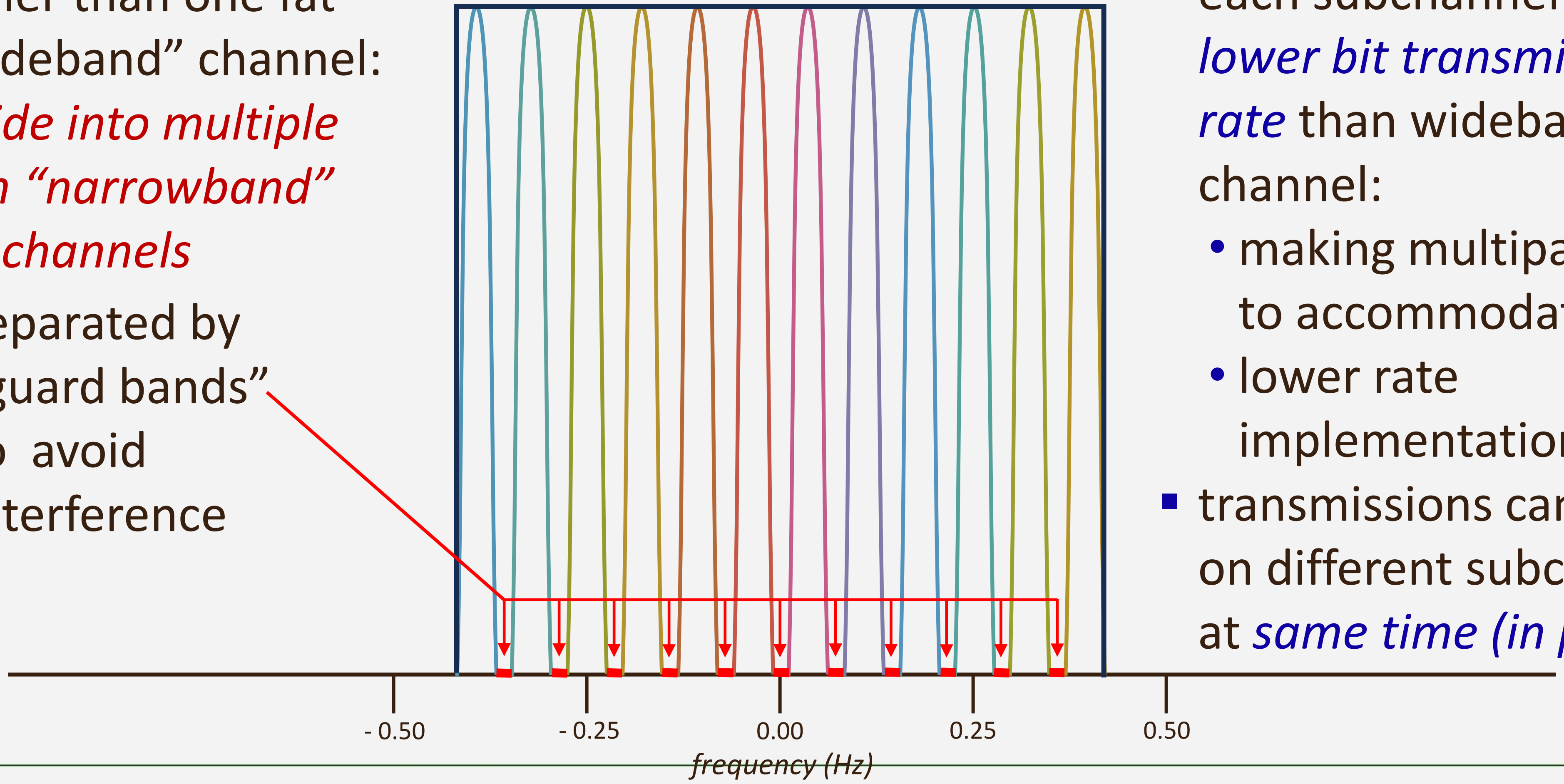
5G RAN

- Sharing the physical radio channel
  - 5G RANs implement OFDMA channel sharing
    - OFDMA combines both frequency multiplexing and time division multiplexing
  - An OFDMA Resource Element (RE)
    - Is a single time minislot at a subcarrier frequency
  - 5G bundles
    - Neighboring REs (in frequency and time) into a Resource Block (RB)
      - A resource block is the smallest unit of channel resource that is used by a transmitting device

FDM: Frequency Division Multiplexing

rather than one fat "wideband" channel: *divide into multiple thin "narrowband" subchannels*

- separated by "guard bands" to avoid interference



Narrowband channels:

- each subchannel has *lower bit transmission rate* than wideband channel:
  - making multipath easier to accommodate
  - lower rate implementations easier
- transmissions can happen on different subchannels at *same time (in parallel)*

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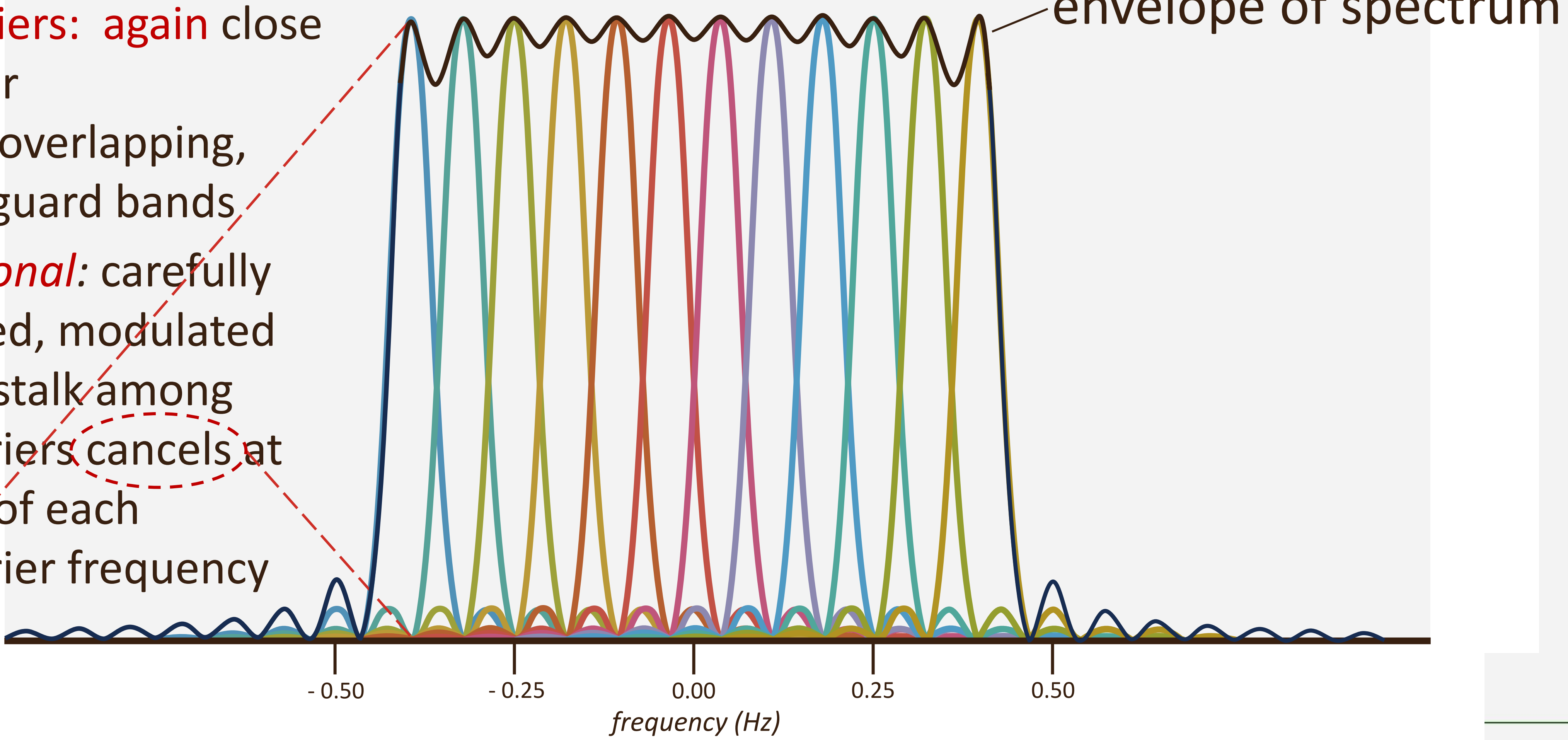
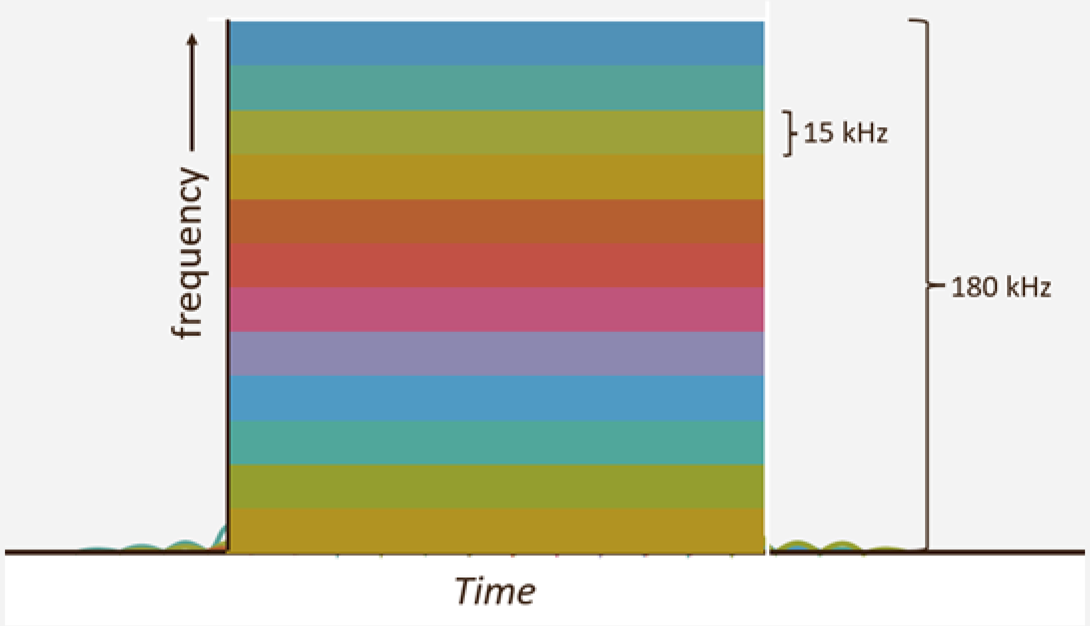
OFDM: *Orthogonal* FDM

- subcarriers: again close together
- slightly overlapping, but no guard bands
- *orthogonal:* carefully designed, modulated so crosstalk among subcarriers cancels at center of each subcarrier frequency



OFDMA: subcarriers



Numerology (μ)	SCS	Symbol Length	Standard Slot Duration
0	15 kHz	≈ 66.67 μ s	1 ms
1	30 kHz	≈ 33.33 μ s	0.5 ms
2	60 kHz	≈ 16.67 μ s	0.25 ms
3	120 kHz	≈ 8.33 μ s	0.125 ms
4	240 kHz	≈ 4.17 μ s	0.0625 ms

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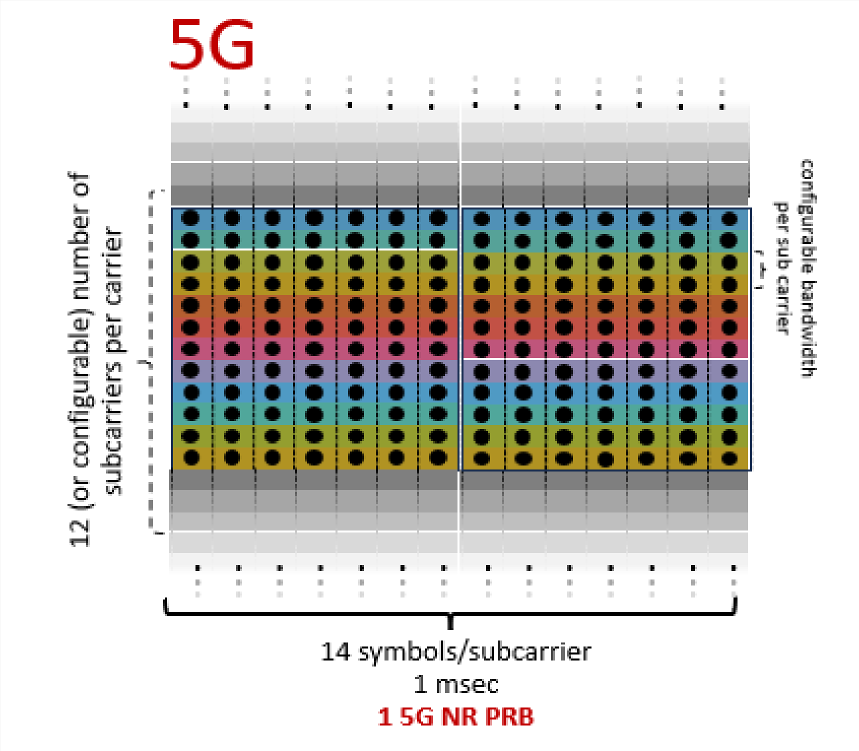
OFDMA: Resource Block

- 5G resource block
  - 12 subcarriers
  - 14 OFDM symbols

- Subcarrier bandwidths:
  - 5G: 15, 30, 60, 120 kHz



5G: resource blocks (RB)

- smallest scheduling unit of transmission to a device

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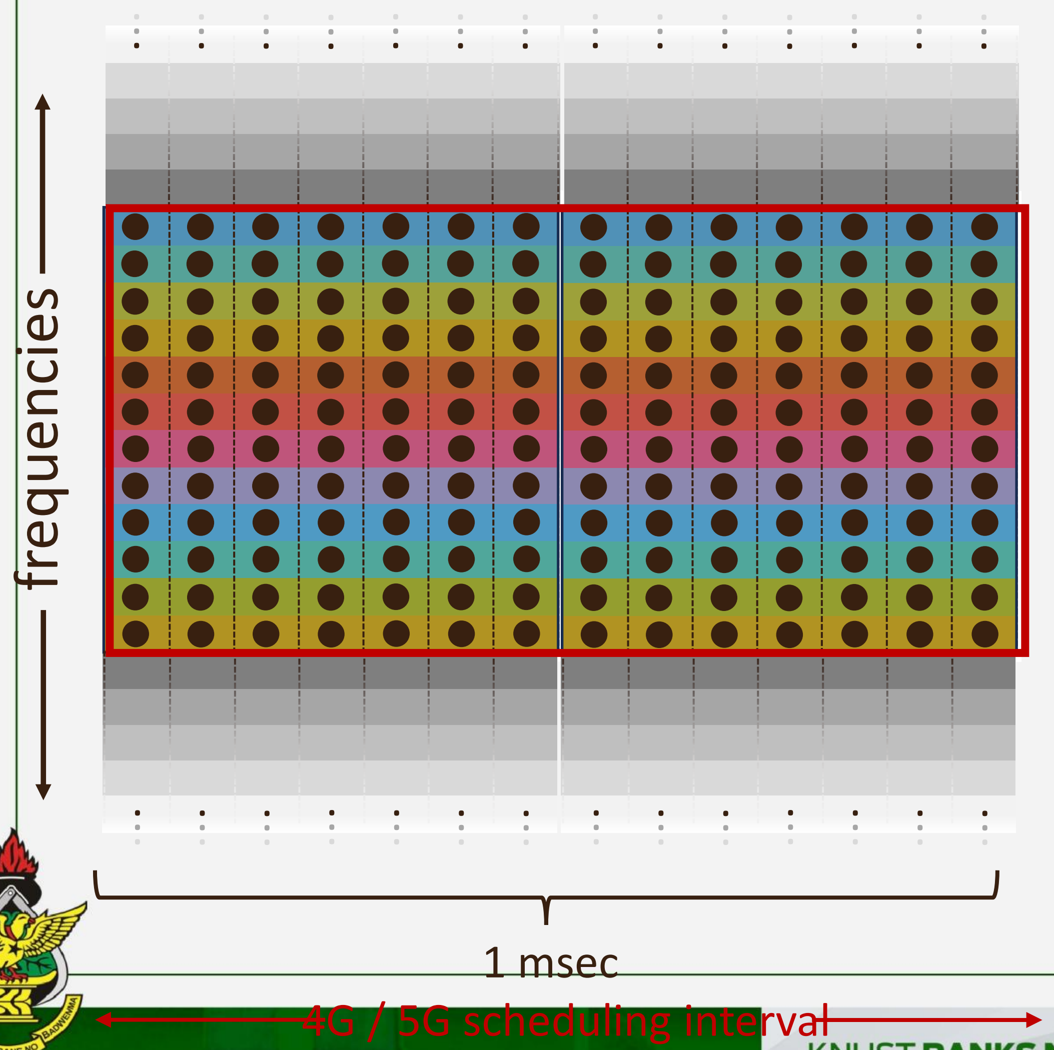
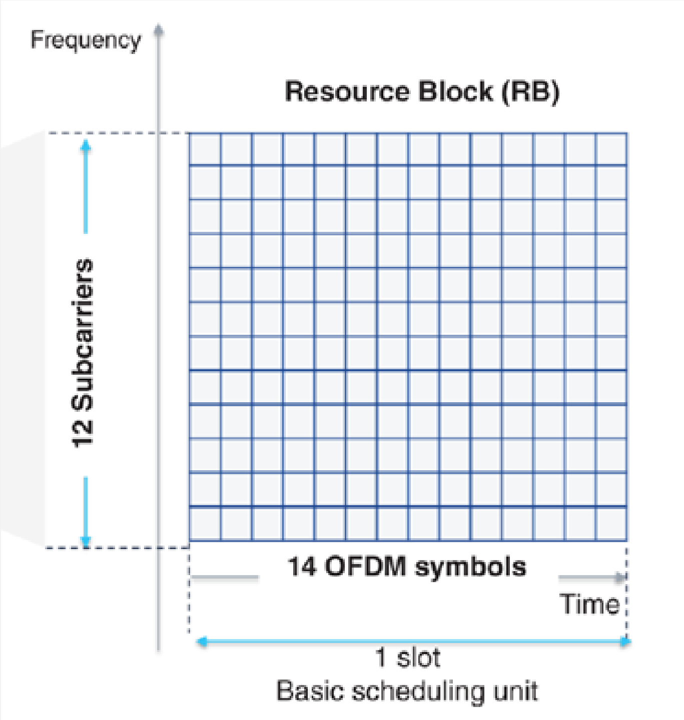
OFDMA: resource block

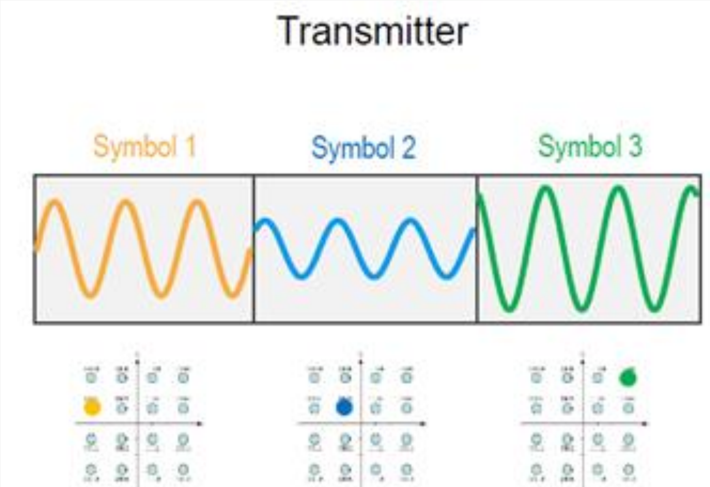


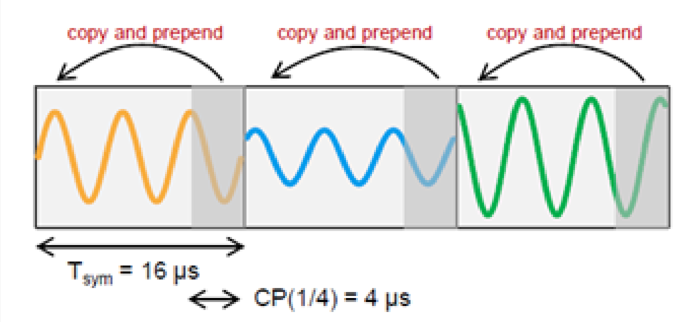
Base station decides which RBs get assigned (over all frequencies) to which devices at 1 msec intervals

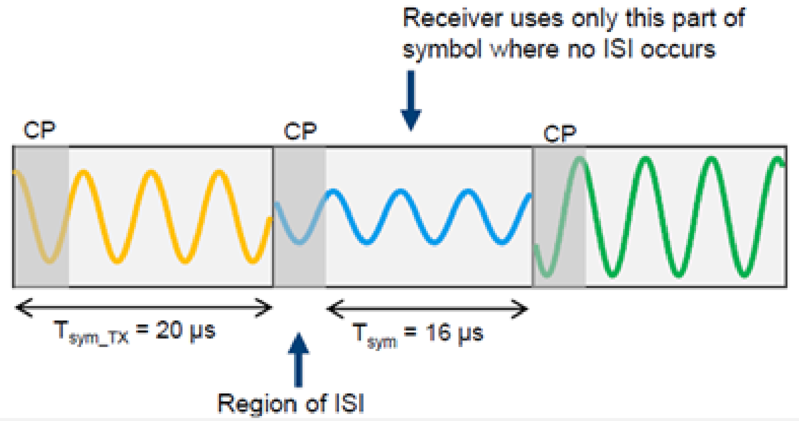


OFDMA

- Visuals







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5G NR PHY Frame Structure

- In 5G NR
  - time is strictly divided into frames, subframes, slots, and symbols
    a. **Radio Frame:** Lasts 10ms. It contains exactly 10 **subframes**
    b. **Subframe:** Lasts 1ms. It is fixed in size regardless of the speed/spacing you use
    c. **Slot:** The basic scheduling unit. A slot holds exactly 14 **OFDMA symbols** (in normal configurations)

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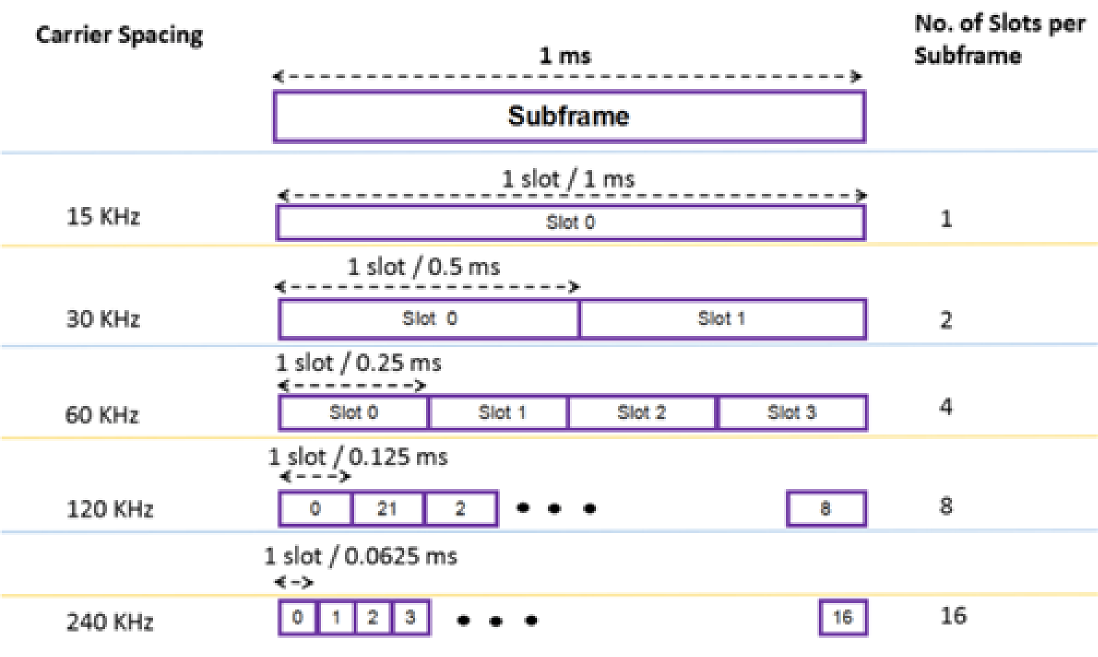
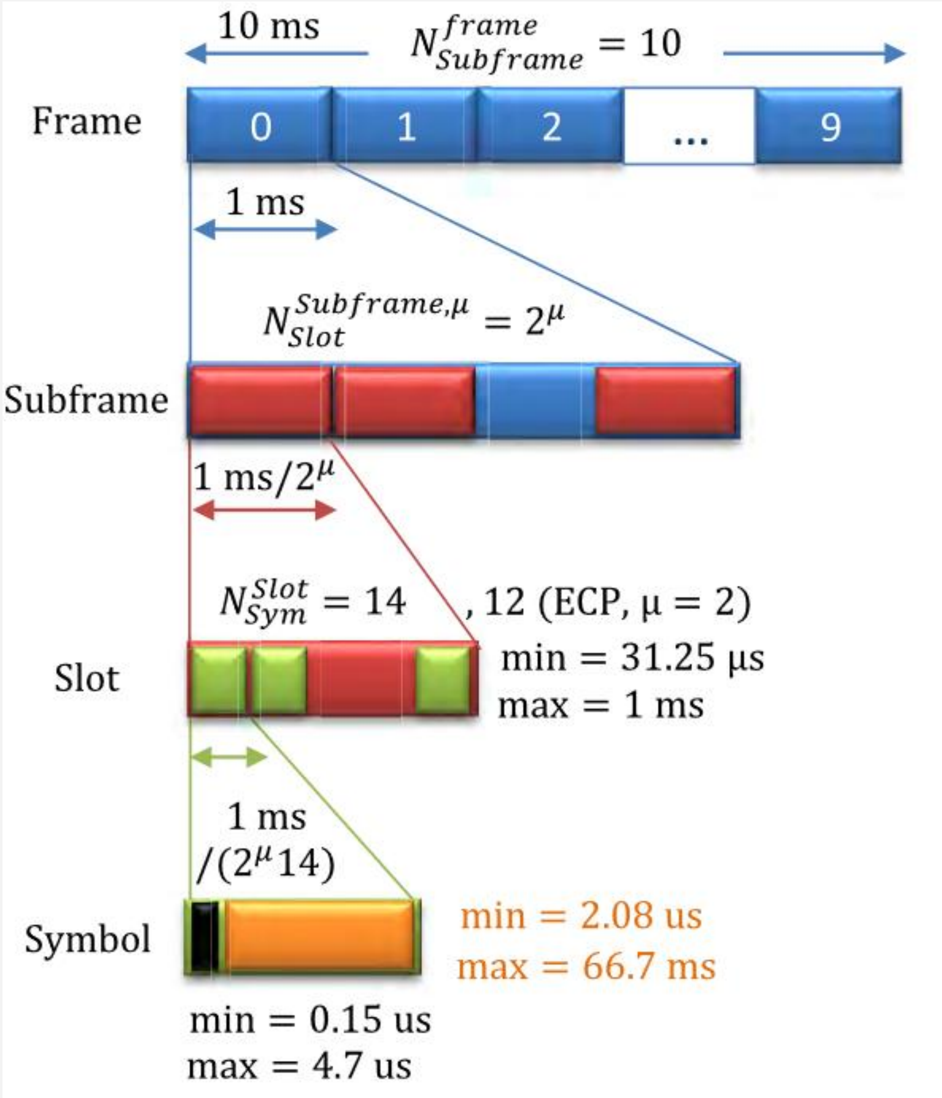
5G NR PHY Frame Structure

- 5G NR
  - utilizes the time units of
    - frame, subframe, slot, and symbol
  - uses the slot for scheduling
  - uses OFDMA symbols for over-the-air transmission





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5G NR PHY Frame Structure

- Time Domain
  - The 5G NR frame structure consists of ten subframes, each of 1ms
  - Thus $N_{subframe}^{frame} = 10$
  - Each 5G NR subframe
    - has a flexible number of slots that depends on the used numerology, μ

$$N_{slot}^{subframe,\mu} = 2^{\mu}$$

$$The\ time\ slot\ period = \frac{1ms}{2^{\mu}}$$

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5G NR PHY Frame Structure

- Time Domain
  - For each slot, there are 14/12 OFDM symbols, denoted by N_{symbol}^{slot} ,for normal/extended cyclic prefix, CP

$$Each\ symbol\ duration = \frac{1ms}{(2^{\mu}\ x\ 14)}\ or\ \frac{1ms}{(2^{\mu}\ x\ 12)}$$

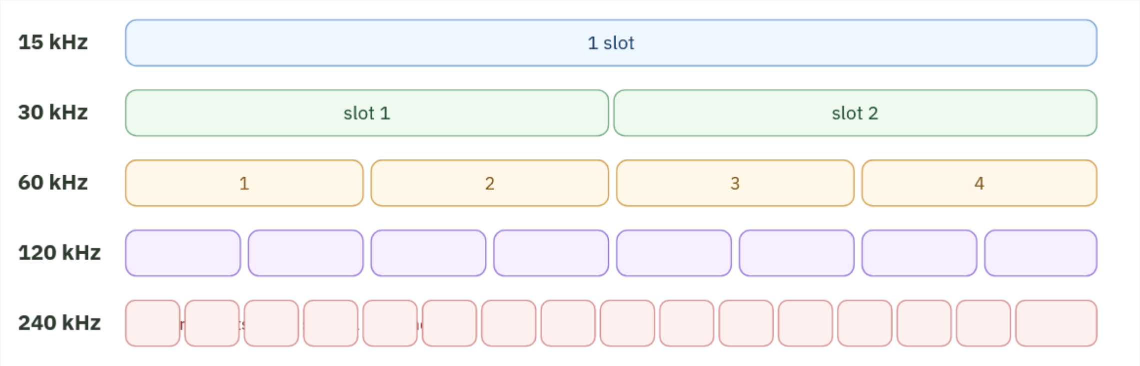
5G NR PHY Frame Structure

- Frequency Domain
  - In the frequency domain, a resource block (RB) is defined as 12 consecutive subcarriers
  - Let Δf denotes the subcarrier spacing
  - In 5G NR
    - Δf is flexible and depends on the used numerology

$$\Delta f = 2^{\mu} \, x \, 15kHz$$

5G NR PHY Frame Structure

- Slots in 1ms subframe



- **μ = 0 (15 kHz):** 1 slot per subframe (1 ms length). Great for
- **μ = 1 (30 kHz):** 2 slots per subframe (0.5 ms length).
- **μ = 2 (60 kHz):** 4 slots per subframe (0.25 ms length).
- **μ = 3 (120 kHz):** 8 slots per subframe (0.125 ms length).
- **μ = 4 (240 kHz):** 16 slots per subframe (0.0625 ms length).

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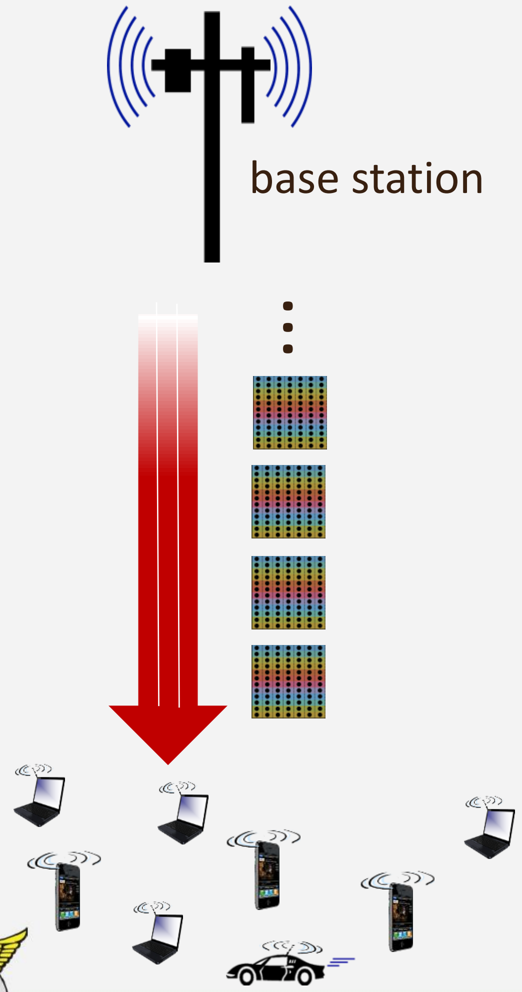
5G Physical channels

- 5G physical layer channels
  - are the actual radio highways that carry data and control signals between a base station and a user device
    - They are split into two groups based on the direction of travel:
      - **Downlink (DL)** (base station to user device)
      - **Uplink (UL)** (user device to base station)
        - Data and signaling messages in NR
          - are carried in the in the DL and UL physical channels

Downlink physical channels



Physical channel

set of Resource Blocks (RBs) carrying similar type of data (channel information RBs, user and control RBs, RB allocation information)

- downlink RBs transmitted from base station to devices
- three 5G downlink physical channels:
  - Physical Downlink Shared Channel (PDSCH)
  - Physical Downlink Control Channel (PDCCH)
  - Physical Broadcast Channel (PBCH)

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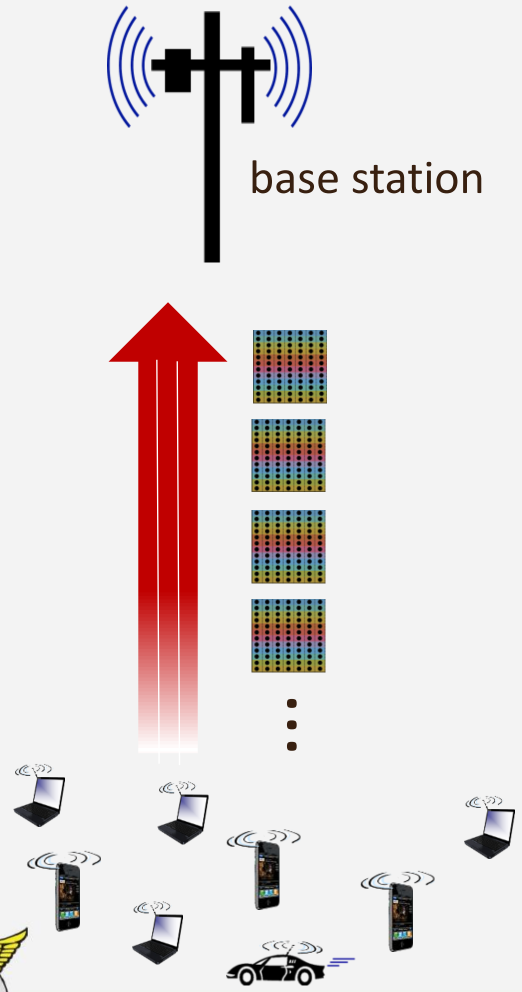
Uplink physical channels



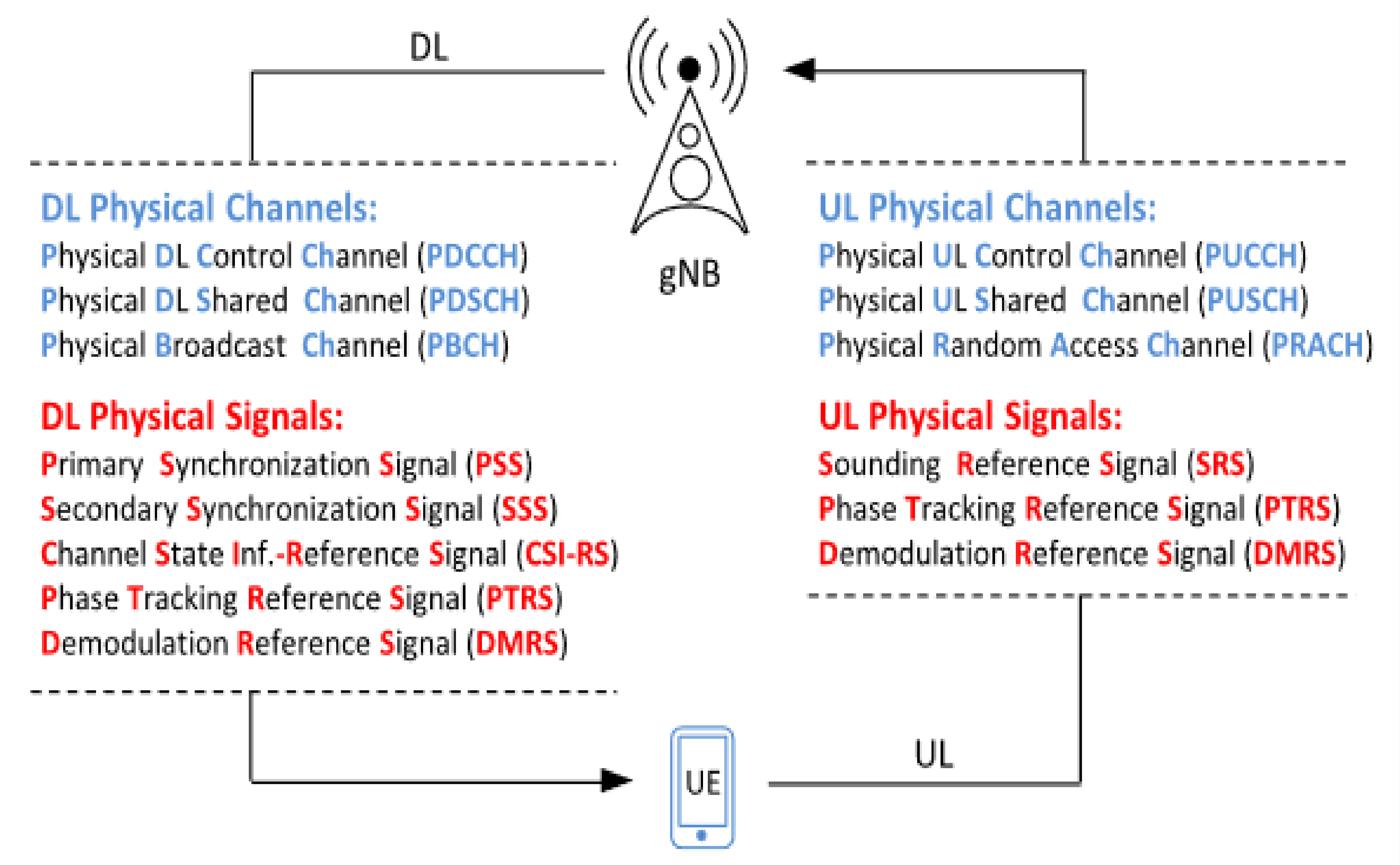
Physical channel

set of Resource Blocks (RBs) carrying similar type of data (radio channel information RBs, user and control RBs, RB allocation information)

- uplink RBs transmitted from devices to base station
- three 5G uplink physical channels:
  - Physical Uplink Shared Channel (PUSCH)
  - Physical Random-Access Channel (PRACH)
  - Physical Uplink Control Channel (PUCCH)

Physical channels and signals

- 5G NR physical channels and signals



Physical channels and signals

- A physical signal corresponds to a set of RE used by
  - the physical layer that does not carry information originating from higher layers
    - These include Synchronization signals (PSS and SSS) or NR reference signals (RS)
- The following physical signals are defined
  - Primary synchronization signal (PSS) and Secondary synchronization signal (SSS):
    - Transmitted together with the PBCH. This pair of downlink signals is used by UE to find, synchronize to, and identify a network

Physical channels and signals

- The following physical signals are defined
  - Demodulation reference signals (DM-RS)
    - for PBCH, PDSCH, PDCCH, PUSCH, and PUCCH:
      - Designed for downlink/uplink channel estimation; coherent demodulation
  - Phase-tracking reference signals (PT-RS):
    - It is a low-density pilot sequence sent at regular time intervals, used to enable tracking of phase noise in both UL/DL

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Physical channels and signals

- The following physical signals are defined
  - Channel-state information reference signal (CSI-RS):
    - UE-specific CSI-RS can be used for estimation of channel-state information (CSI) to further prepare feedback reporting to gNB to assist in MCS selection, beamforming, MIMO rank selection, and resource allocation
  - Tracking reference signals (TRS):
    - Time and frequency TRS is designed for time/frequency tracking and estimation of delay/Doppler spread
      - TRS is configured as a CSI-RS with specific parameter restrictions (time/frequency location, RE pattern, etc.)

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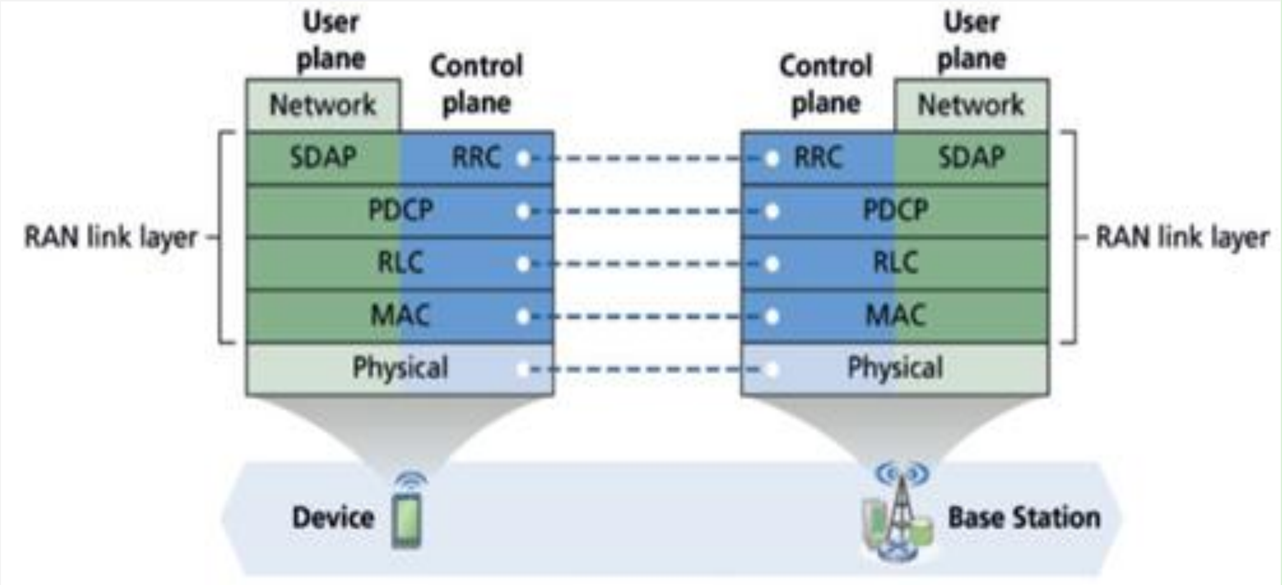
Physical channels and signals

- The following physical signals are defined
  - Sounding reference signal (SRS):
    - UE-specific SRS can be used for estimation of uplink CSI to assist uplink scheduling, and uplink power control, as well as assist the downlink transmission
      - e.g. the downlink beamforming in the scenario with UL/DL reciprocity

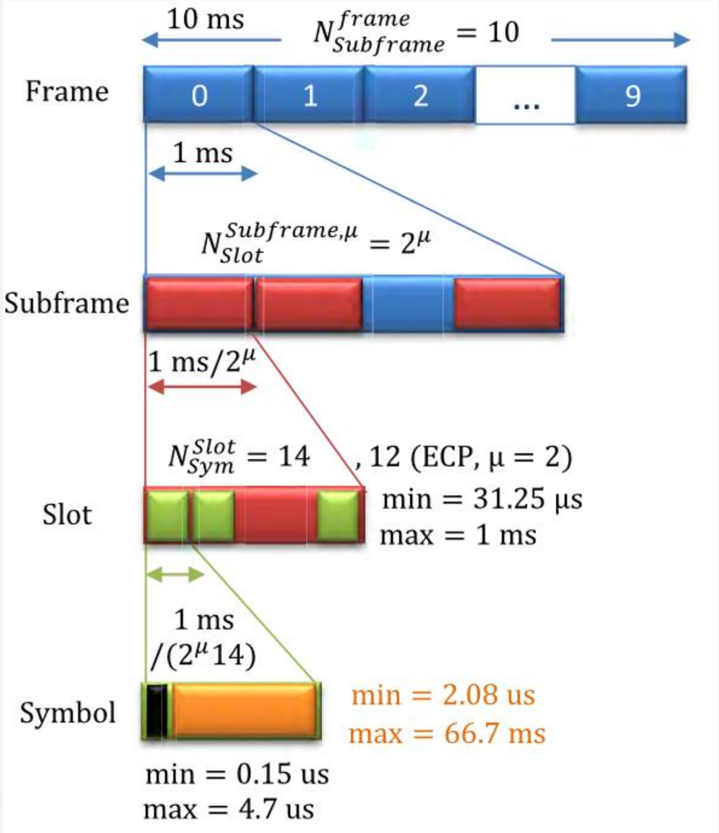
5G NR Protocol Stack

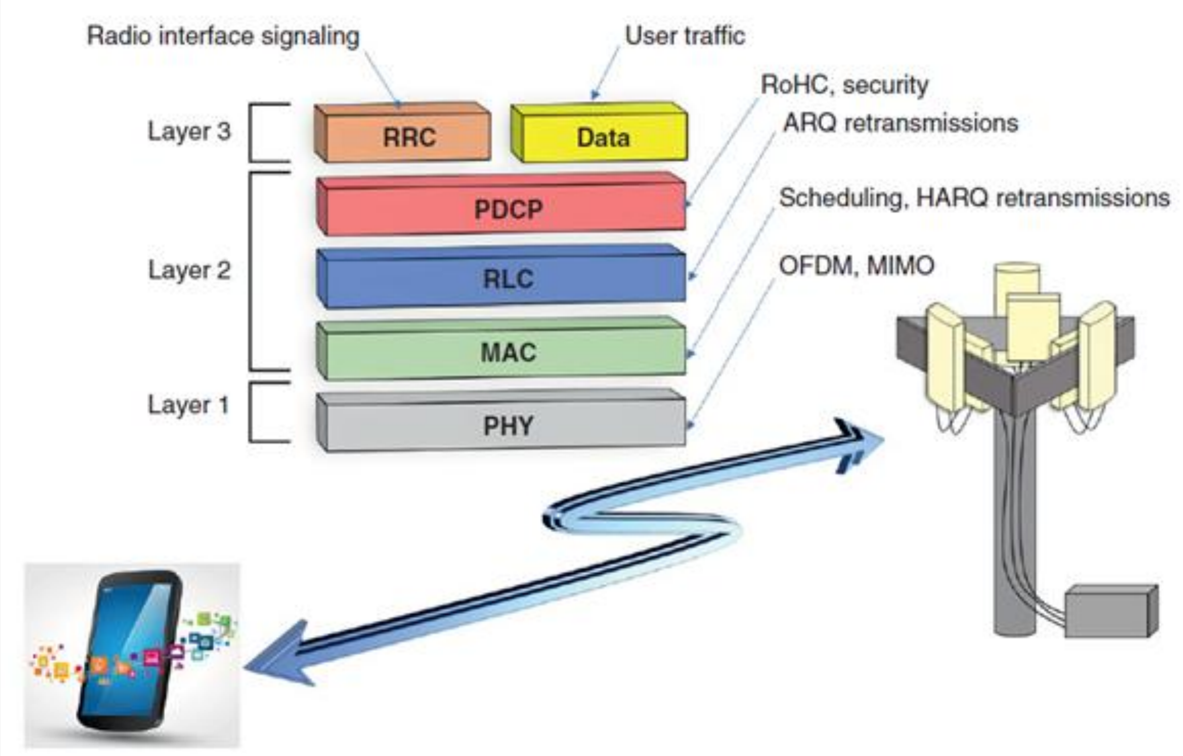
- It is a layered system that
  - manages how data travels between a user device and the network (gNB)
    - It efficiently handles both signaling messages and user data



5G NR Protocol Stack

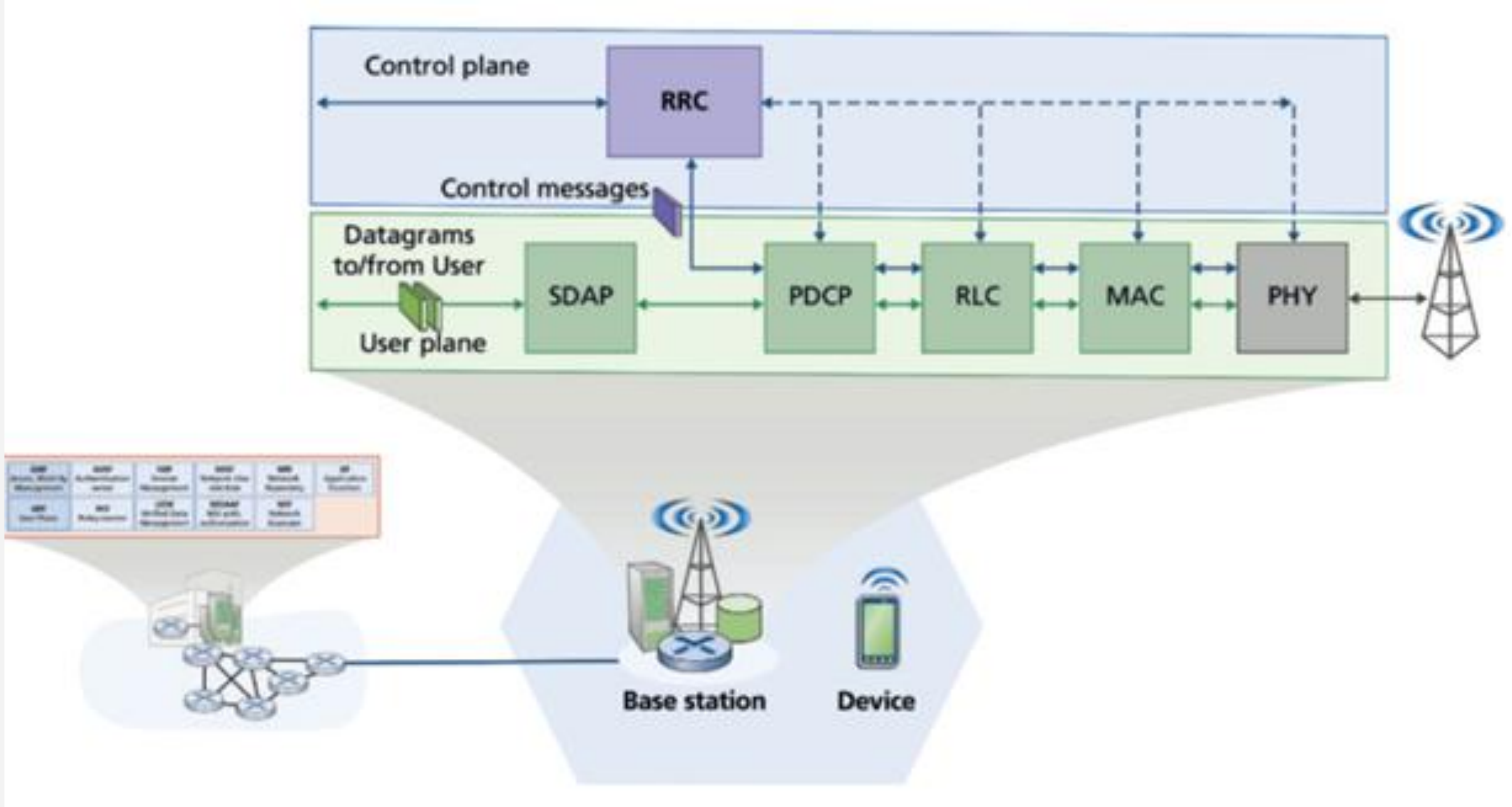
- The frame structure along with the radio interface protocol stack
  - enables the UE and the radio network to communicate successfully with each other





5G NR packet processing pipeline

- It is split into a user plane and a control plane



5G NR Protocol Stack

- The job of the user plane is
  - to move link-layer frames from the device's applications to the base station and *vice versa*
- The job of the control plane is
  - to control and configure the RAN's radio resources
    - provide control functions for user-plane frame transfer (e.g., ARQ and quality of service), and additional services such as device mobility, attachment to the network, and power control

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5G NR Protocol Stack

- Radio Resource Control (RRC)
  - Ensures that
    - cells get connected
    - Data flows smoothly
    - Users stay seamlessly connected even on the move
  - Thus it carries access stratum (AS) signaling
    - such as mobility management-related signaling and configuration of other protocols

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5G NR Protocol Stack

- User traffic
  - enters the radio protocol stack as a Layer 3 information
- SDAP
  - facilitates the implementation of 5G QoS
    - SDAP assigns QoS Flow IDs and maps the flows onto underlying radio resources
- The PDCP functions
  - include Robust Header Compression (RoHC) and security mechanisms such as ciphering and integrity protection
    - It also adds new functions, such as support for integrity protection for the user plane and duplication of signaling and user traffic for enhanced reliability

5G NR Protocol Stack

- The RLC protocol
  - supports transparent, unacknowledged, and acknowledged modes
    - Additionally, 5G NR's RLC protocol supports variable length TTIs resulting from different OFDM numerologies that utilize different SCS values
- The MAC protocol
  - supports multiplexing of RLC PDUs, prioritization among and within UEs, and scheduling
    - Additionally, 5G NR has a more flexible HARQ that supports Code Block Group-based HARQ

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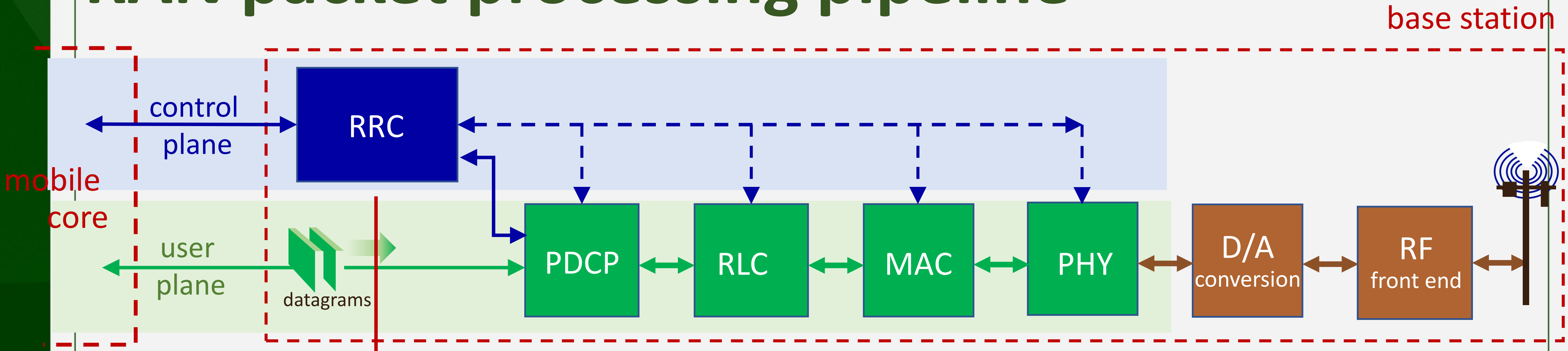
5G NR Protocol Stack

- The PHY layer
  - implements baseband and RF processing.
    - For example, PHY layer functions are OFDM processing and MIMO processing
      - 5G NR utilizes new coding techniques of polar coding and LDPC coding to increase spectral efficiency

RAN packet processing pipeline

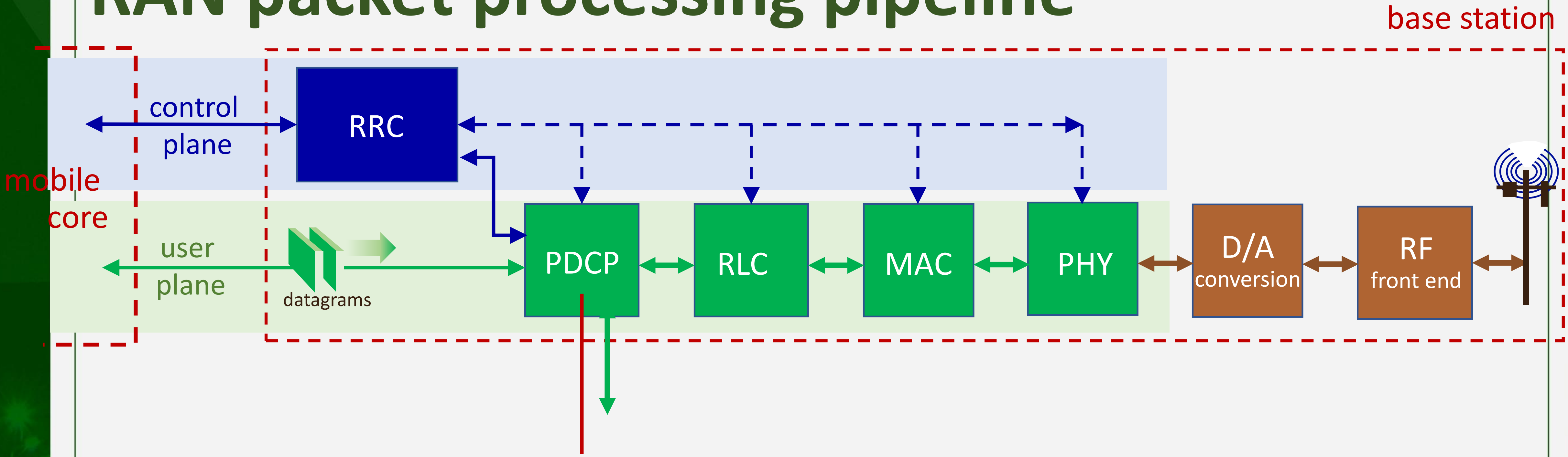


traditional 3GPP view

RRC (Radio Resource Control)

- configures coarse-grained, policy-related aspects of pipeline (e.g., scheduling prioritization, security)
- this implements the RAN's control plane
- does not process user plane packets

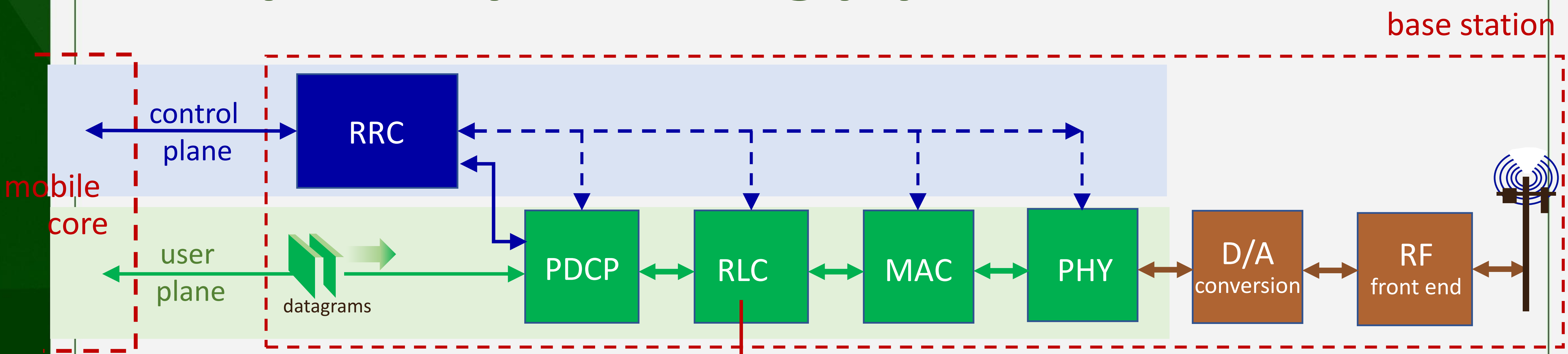
RAN packet processing pipeline



PDCP (Packet Data Convergence Protocol)

- compress/decompress IP headers
- ciphering, integrity protection

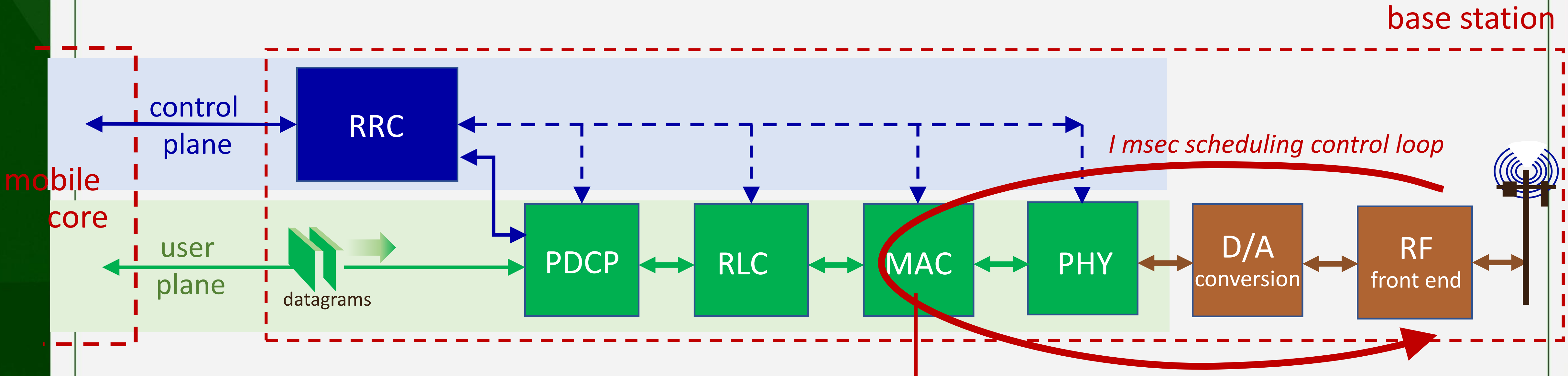
RAN packet processing pipeline

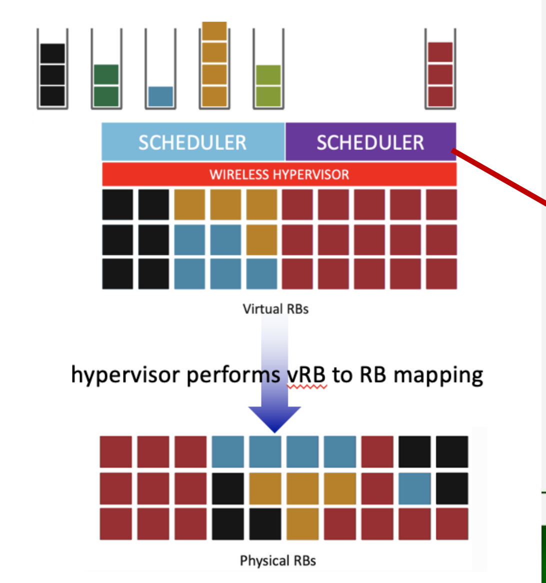


RLC (radio link control)

- link-layer frame segmentation/reassembly
- reliable data transfer (ARQ)

RAN packet processing pipeline

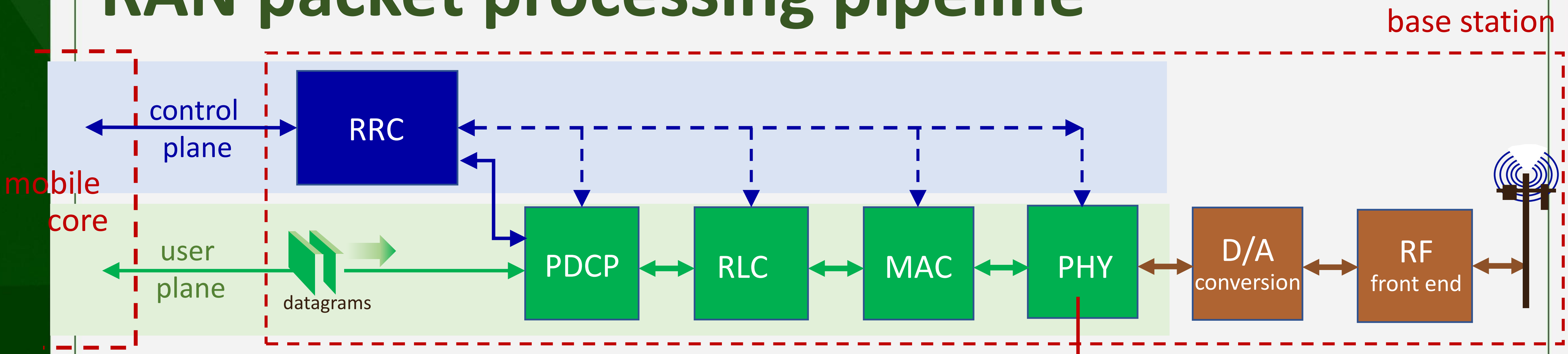




MAC (Media Access Control)

- buffering, multiplexing/demultiplexing frames
- scheduling: real-time scheduling frame decisions

RAN packet processing pipeline

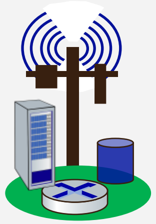


PHY (Physical Layer)

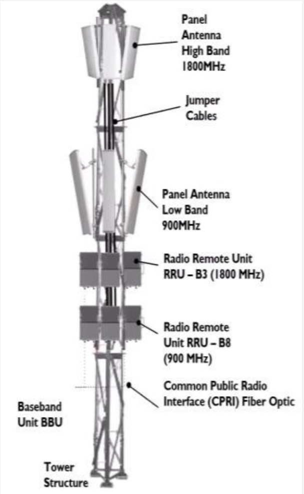
- modulation, coding

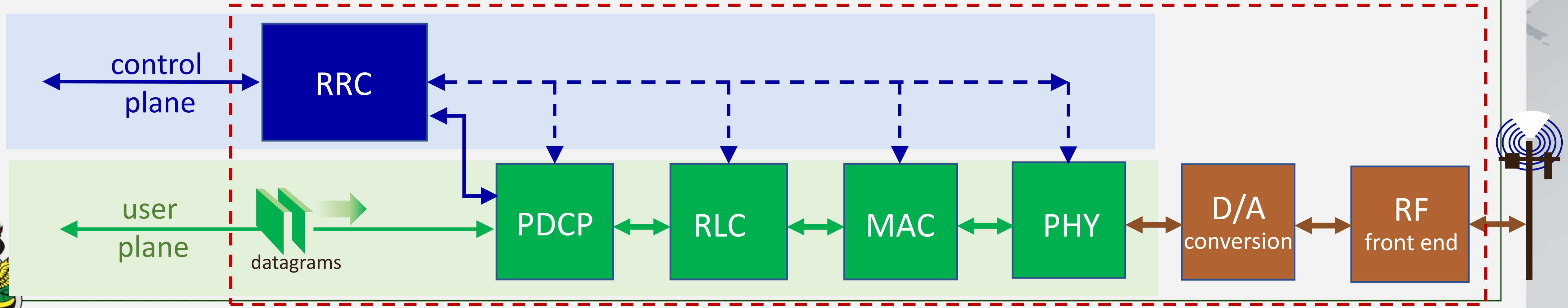
Split RAN

- How is RAN *functionality* partitioned between physical elements (i.e.,., “split” across centralized and distributed locations)?
  - historically: no split - implemented in base station



Monolithic base station





Split RAN

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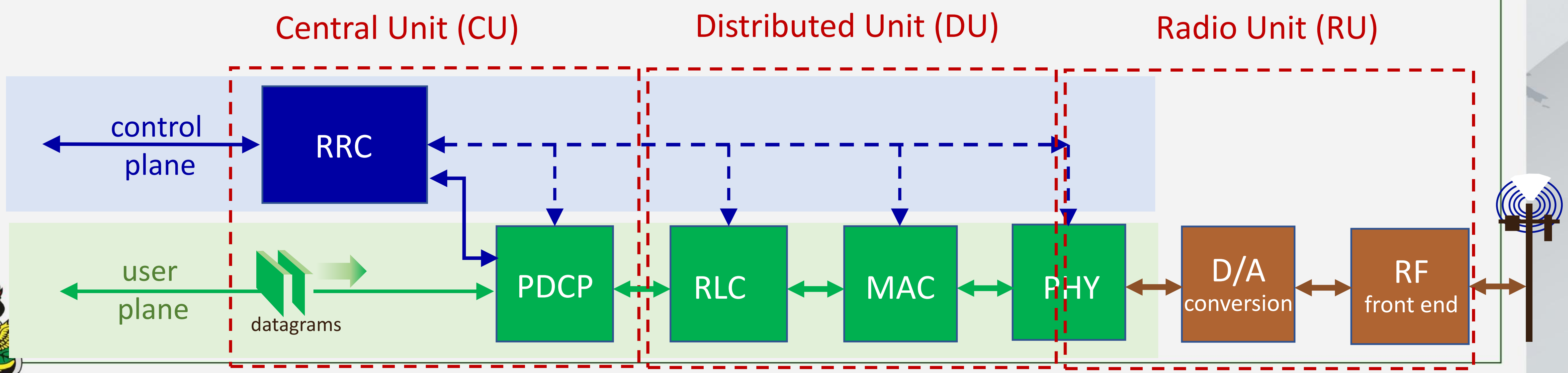
  - O-RAN: three "units" - *where* units are implemented is an *implementation* choice



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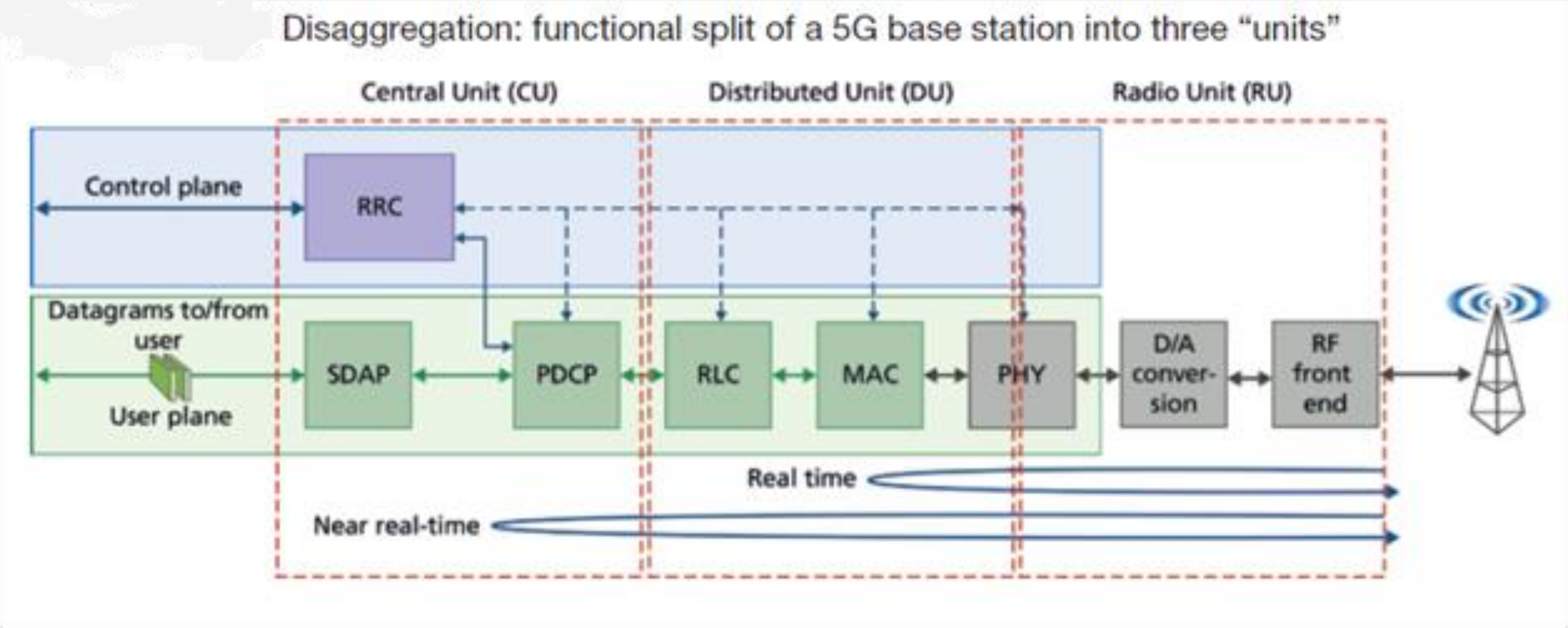


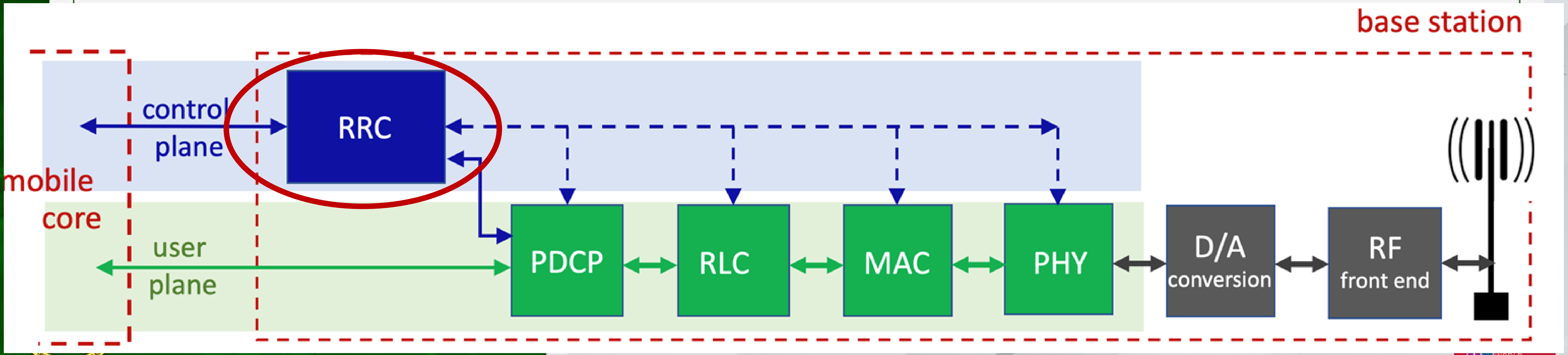
Split RAN

- **The Centralized Unit (CU)**
  - It is responsible for managing and controlling radio resources, network planning, and user mobility
- **The Distributed Unit (DU)**
  - focuses on real-time radio signal processing and the management of the physical layer functions, by ensuring low-latency operations thanks to the proximity to the UE

Software-defined RAN

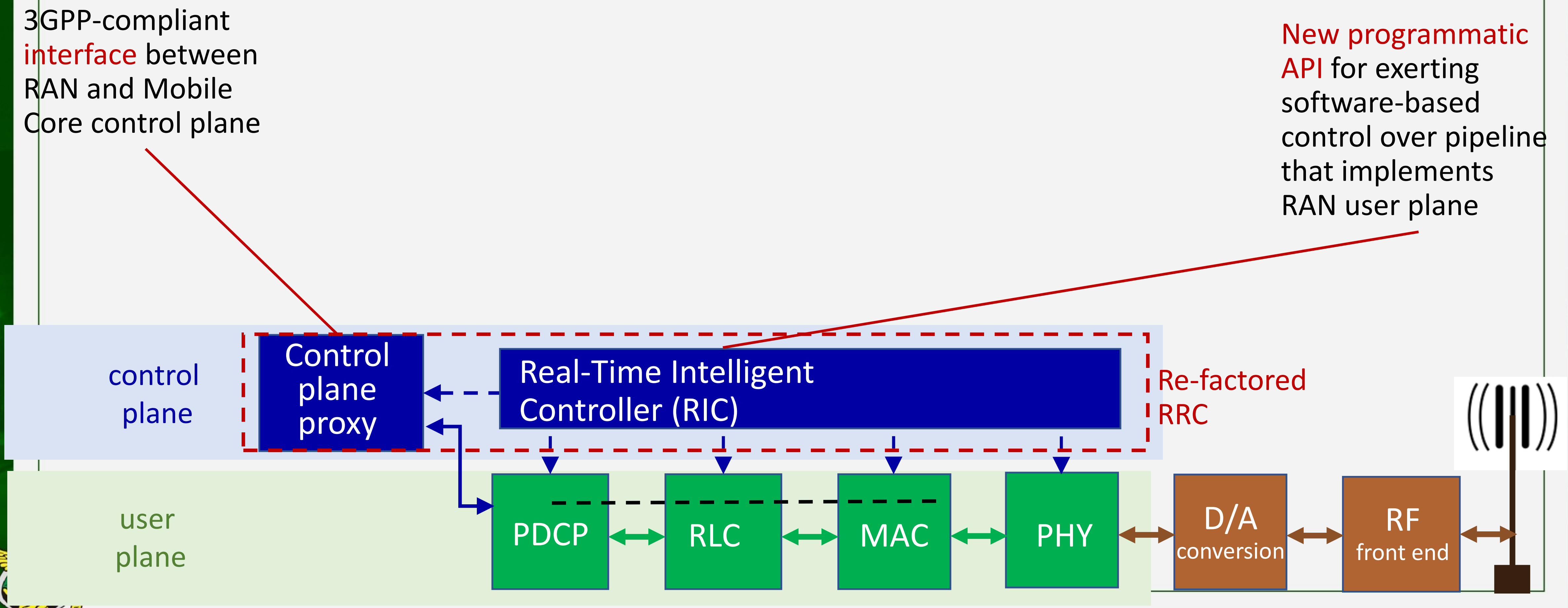
Recall our earlier description of traditional RAN base station (below)

- tightly coupled control and data planes
- let's focus on control / management: RRC *implementation*



Software-defined RAN

- *SD-RAN:* implementing RAN using SDN approach



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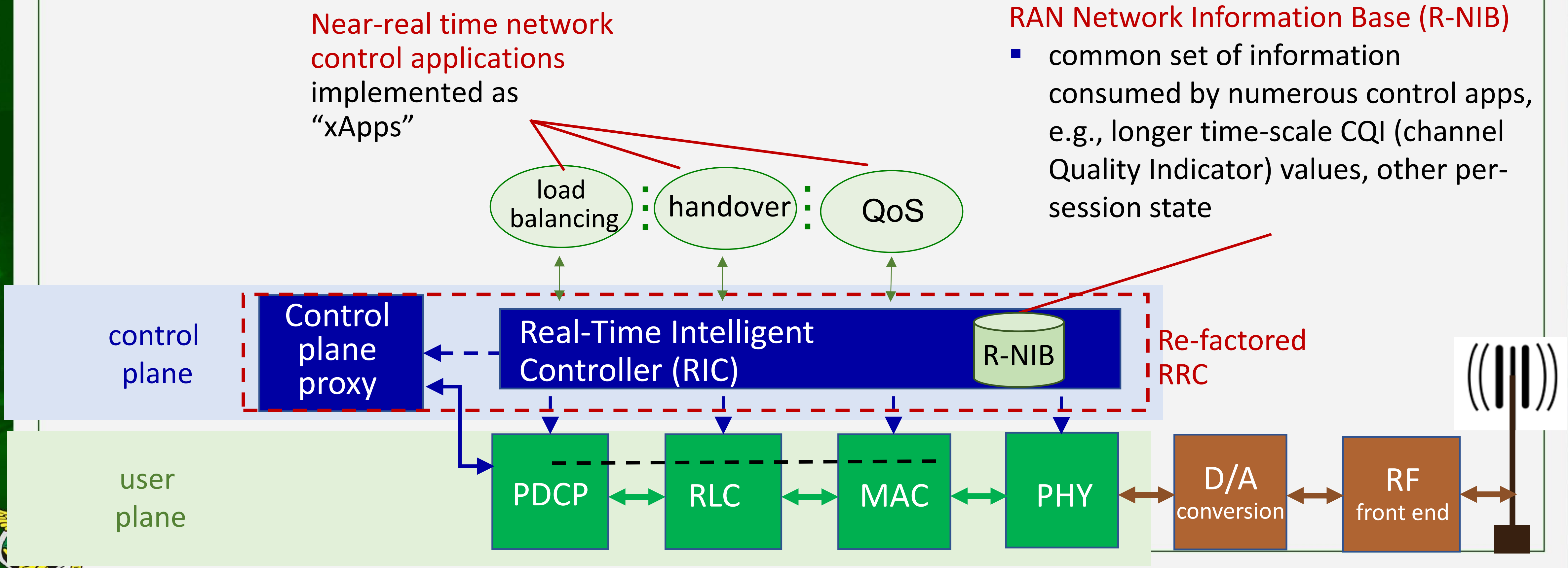
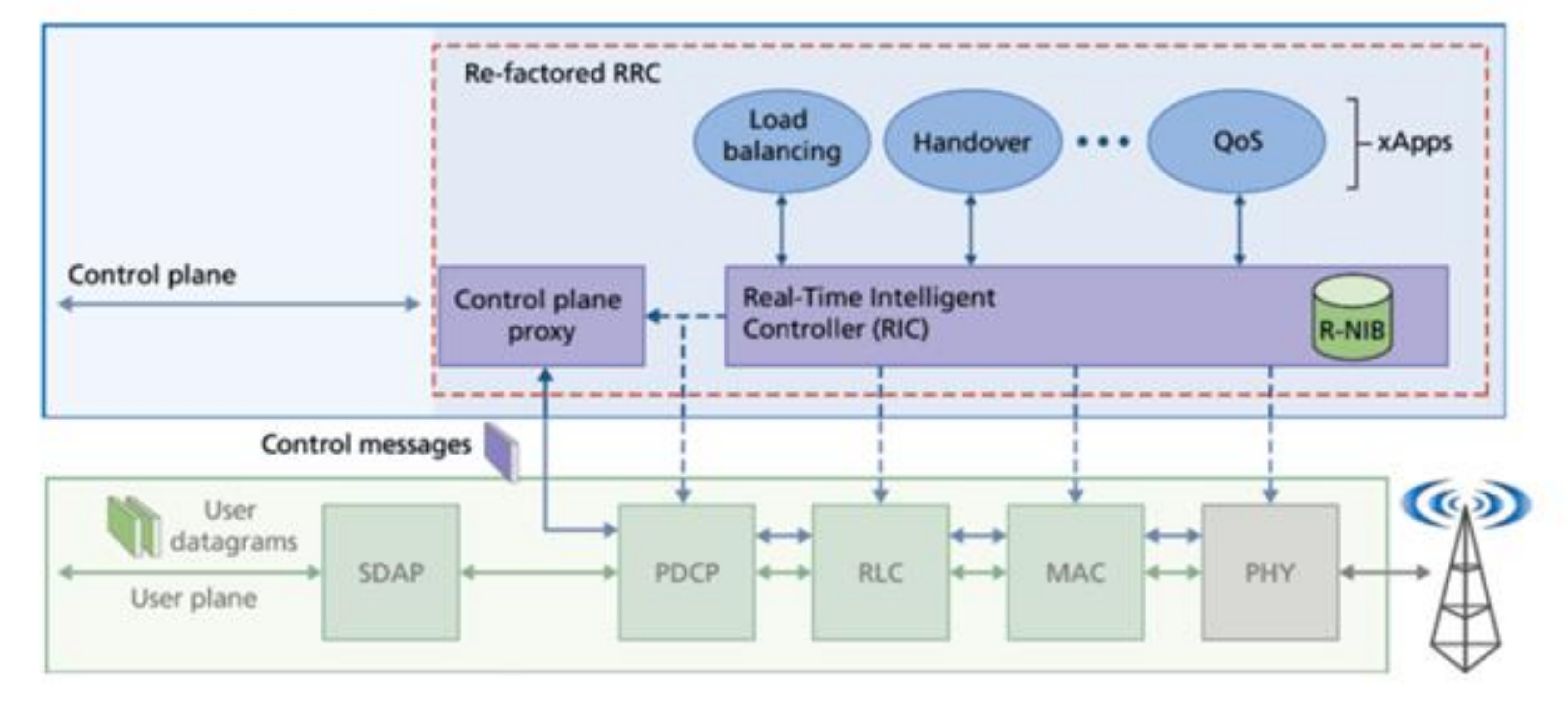
Software-defined RAN

- *SD-RAN:* implementing RAN using SDN approach



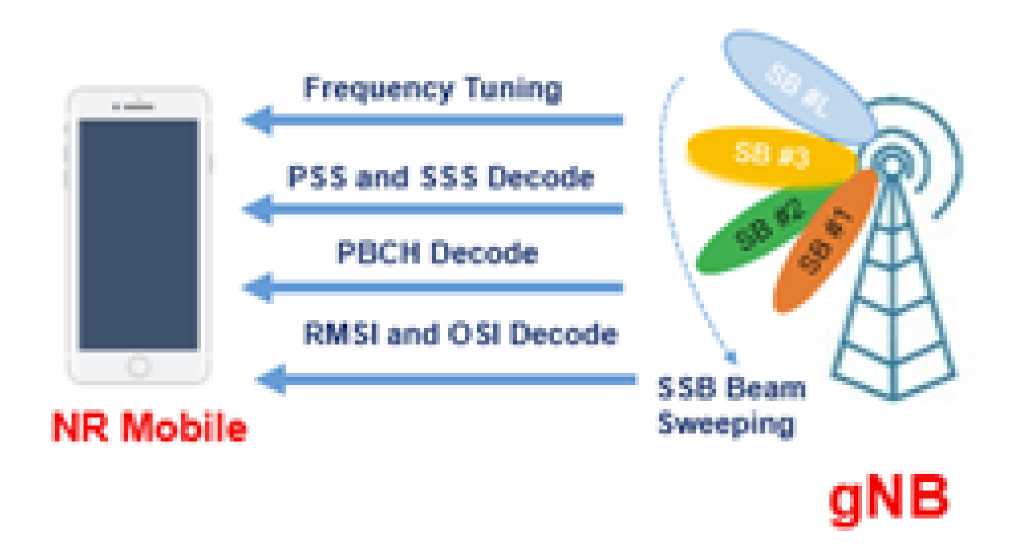
Software-Defined RAN

- Software Defined RAN



5G Initial Access Procedure

- The Initial Access procedure
  - Describes how a UE discovers a cell, synchronize with it, and establishes an initial connection with the gNB
    - The process is beam-based



- Thus, the process involves three main phases
  1. Cell search (finding the signal)
  2. System information reading (learning network rules)
  3. Random access (registering the UE with gNB)

5G Initial Access Procedure

1. Synchronization Signals (PSS / SSS)
   - The gNB periodically transmits PSS and SSS on different beams
   - The UE scans the NR carrier and searches for these signals
     - UE Actions:
       - Detects the cell
       - Achieves time and frequency synchronization
       - Identifies the best beam based on signal quality
   - This step allows
     - the UE to synchronize with the gNB and prepare for system information decoding

5G Initial Access Procedure

2. PBCH – Master Information Block (MIB)

- The gNB transmits PBCH, which carries the MIB
- The MIB is transmitted within the SS Block (SSB)
  - UE Actions:
    - Decodes the MIB on the selected beam
    - Obtains essential information such as:
      - System frame number
      - Subcarrier spacing
      - Scheduling information for SIB1

5G Initial Access Procedure

3. RMSI (SIB1) and Other System Information

- After MIB, the UE decodes RMSI (Remaining Minimum System Information)
- This includes SIB1, which provides:
  - Cell access parameters
  - PLMN information
  - RACH configuration

- At this stage
  - The UE has enough information to attempt Random Access

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5G Initial Access Procedure

4. Random Access Procedure (RACH)
   - Message1 – RACH Preamble
     - UE transmits a PRACH preamble on the configured RACH resource
     - The preamble is beam-based and aligned with the selected SSB
   - Message2 – Random Access Response (RAR)
     - gNB responds with RAR, which contains:
       - Timing Advance (TA)
       - Temporary Cell Radio Network Temporary Identifier (C-RNTI)
       - Initial uplink grant
   - This step synchronizes the UE uplink with the gNB

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5G Initial Access Procedure

4. Random Access Procedure (RACH)

- Message 3 – UE Transmission
    - UE sends Msg3 using the granted uplink resources
        - Msg3 typically carries the RRC Connection Request
- Message 4 – gNB Response
    - gNB responds with Msg4, such as:
        - RRC Connection Setup
- After this step
    - the UE moves toward RRC Connected state

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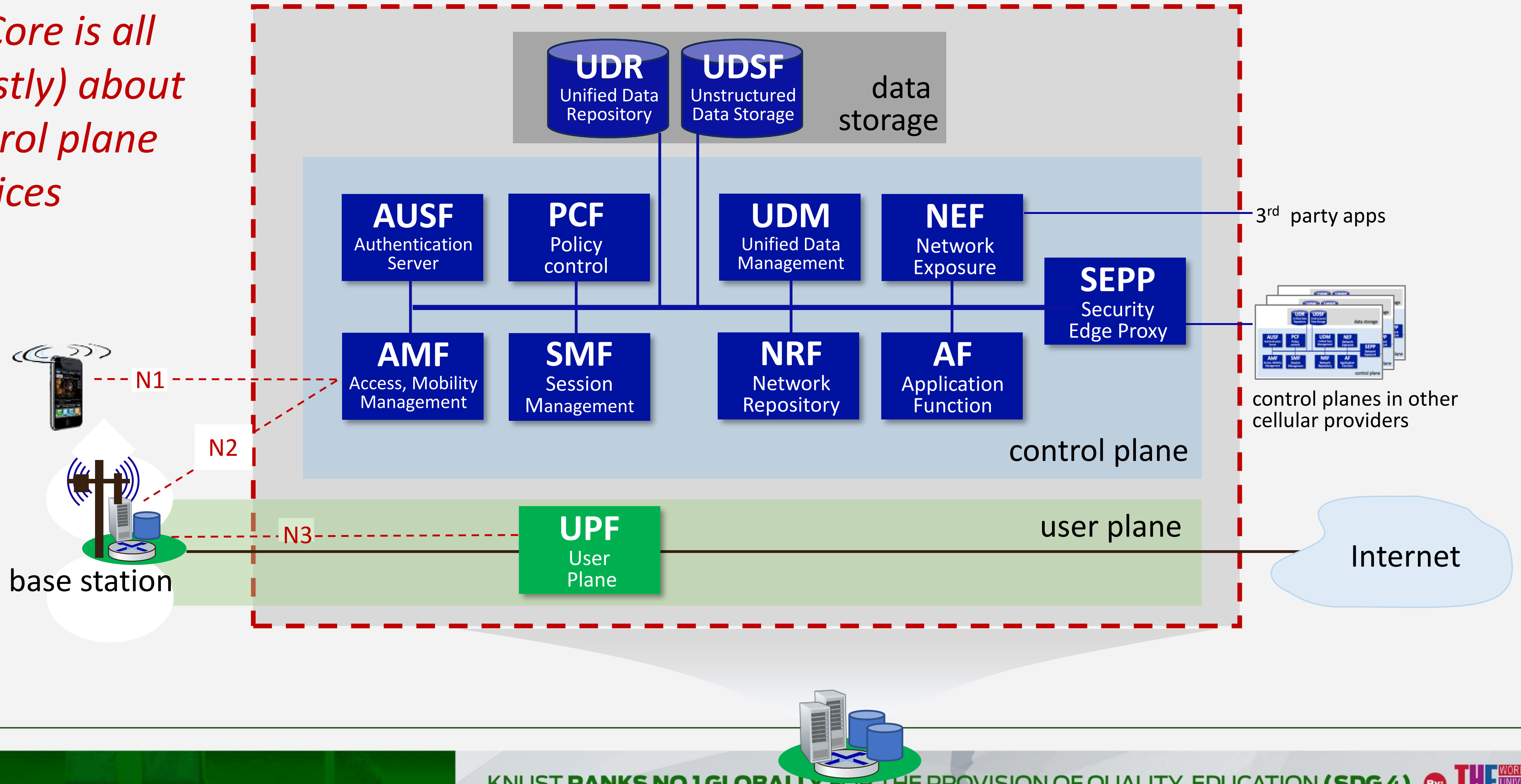
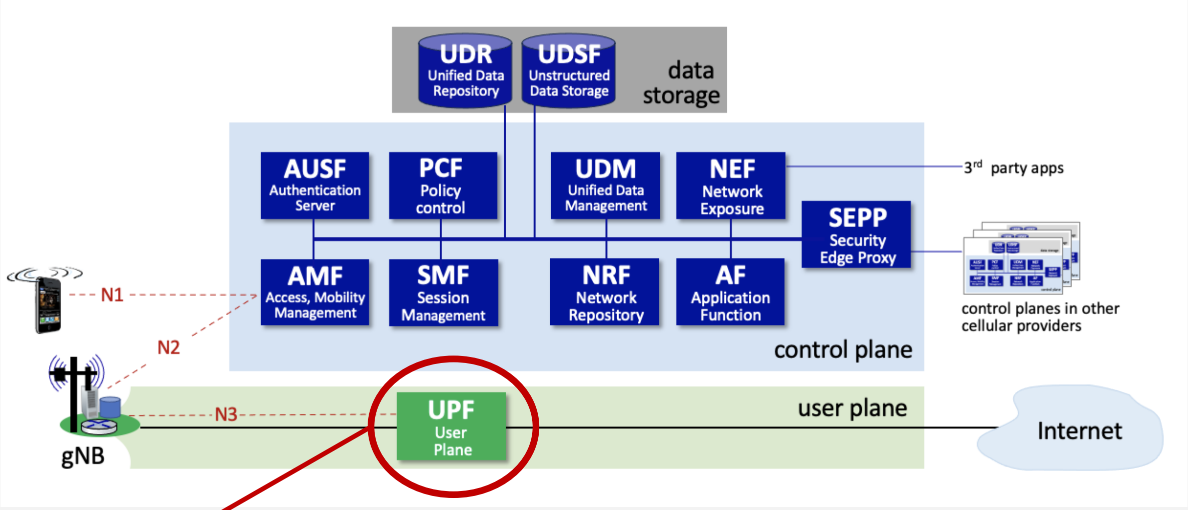
5G Core Functional Components

5G Core is all (mostly) about control plane services



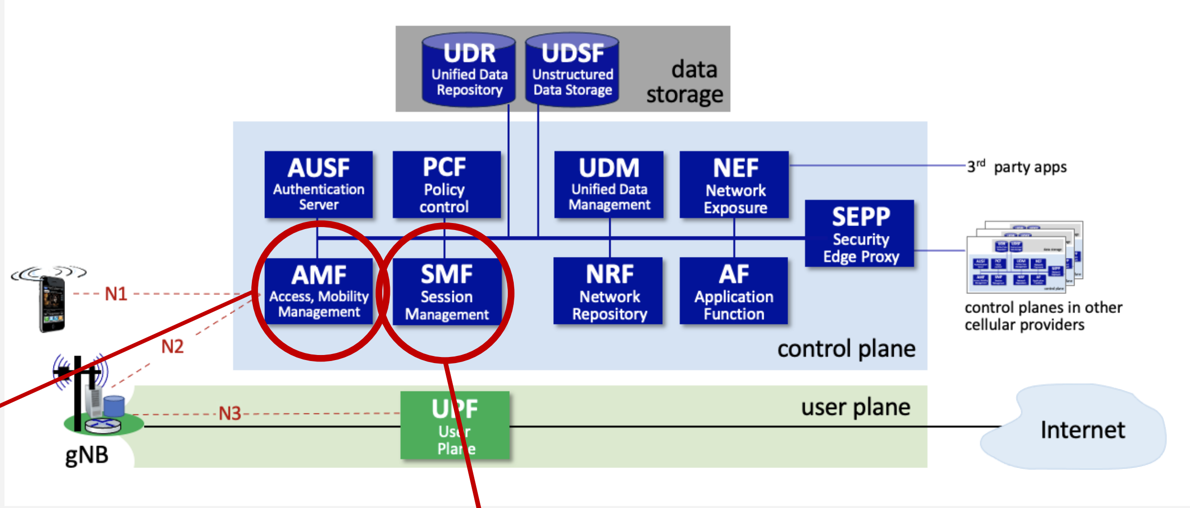
5G Core Functional Components



UPF (User Plane Function)

- each device is assigned to a UPF, which acts as relay, forwarding traffic between UE, Internet
- UPF responsible for policy enforcement, lawful intercept, traffic usage measurement, QoS policing
- responsible for tunneling (i.e., encapsulating, decapsulating) traffic to/from base station over N3 interface

5G Core Functional Components



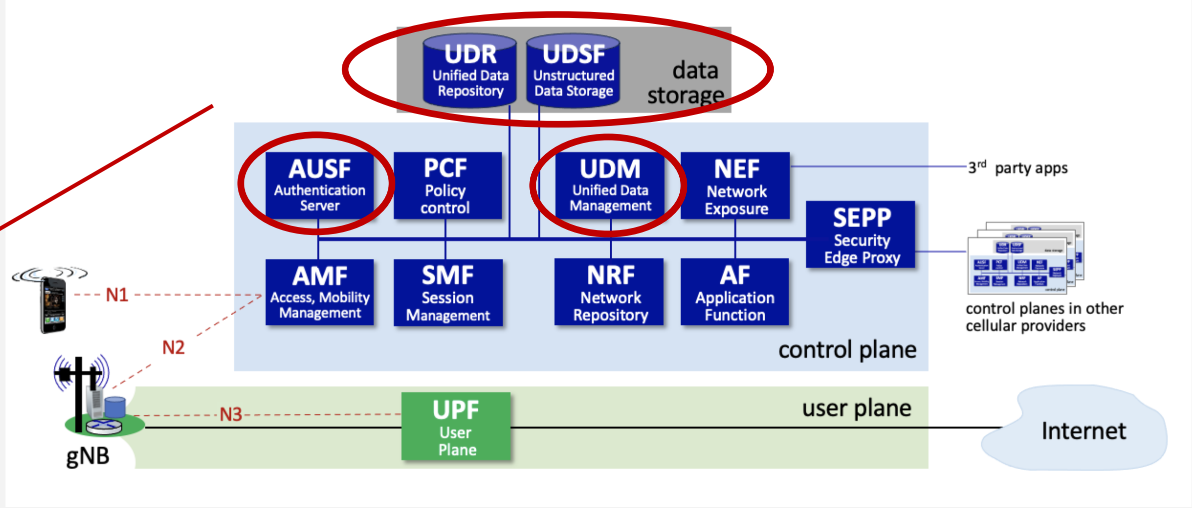
AMF (Access, Mobility Management Function)

- connection, reachability management; mobility management; access authorization; and location services

SMF (Session Management Function)

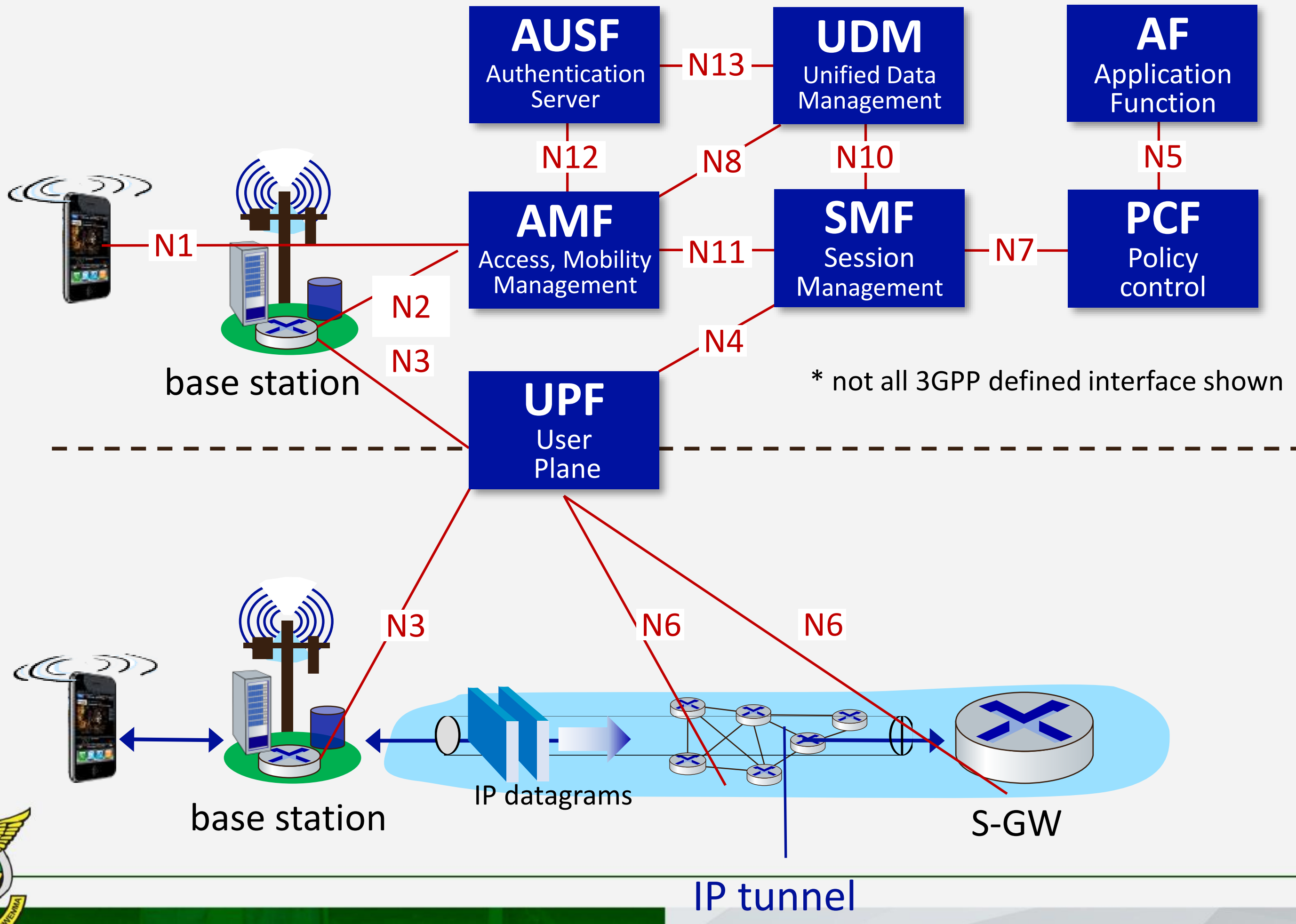
- manages each UE session, including IP address allocation, selection of associated UP function, QoS control, control aspects of UP routing

5G Core Functional Components



- *AUSF (Authentication Server Function):* Authenticates devices.
- *UDM (Unified Data Management):* Manages user identity, including the generation of authentication credentials.
- *UDR (Unified Data Repository):* Manages user static subscriber-related information.
- *UDSF (Unstructured Data Storage Network Function):* Used to store unstructured data, and so is similar to a *Key/Value Store*.

5G interfaces*, control/user -plane separation (CUPS)



control plane

- emphasis on functions and services
- well-defined interfaces defines between services

data plane

- 5-layer Internet data plane
- extensive use of tunneling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